Self-Adaptive Improved Fungal Growth Optimization: A Hybrid Metaheuristic Framework for High-Quality, Robust and Scalable Clustering

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Abstract

Clustering is a fundamental task in machine learning and data science. Traditional clustering algorithms are highly sensitive to initialization and are often prone to becoming trapped in local optima, particularly when dealing with complex or high-dimensional datasets. To overcome these limitations, this article proposes a *Self-Adaptive Improved Fungal Growth Optimization* (SAIFGO) algorithm, an enhanced version of the nature-inspired *Fungal Growth Optimization* (FGO) algorithm. The proposed SAIFGO algorithm mimics key fungal behaviors, including hyphal growth, chemotropism, branching, and spore germination. It incorporates several advanced strategies, such as chaotic initialization, Lévy flight-based mutation, opposition-based learning, and an adaptive exploration–exploitation mechanism, to improve the search process. Furthermore, a composite multi-objective fitness function based on multiple cluster validity indices is employed to achieve superior clustering performance. The proposed algorithm is also hybridized with conventional clustering techniques, including K-Means and Fuzzy C-Means (FCM), to optimize cluster centroids more effectively. Extensive experiments are conducted by comparing SAIFGO with several state-of-the-art metaheuristic algorithms using five cluster validity indices and comprehensive statistical analyses. The experimental results demonstrate that SAIFGO consistently outperforms or remains highly competitive with existing methods across

diverse datasets. These findings indicate that SAIFGO is an effective, robust, and high-performing optimization algorithm for solving complex data clustering problems.

Keywords: K-Means, FCM, Fungal Growth Optimization (FGO), Self-Adaptive Improved Fungal Growth Optimization (SAIFGO)

1 Introduction

Clustering [1] is a fundamental unsupervised learning method for identifying inherent structures present within unlabeled data. It is a vital component in various applications, including pattern recognition [2], image segmentation [3] and data analysis where it simplifies complex data to facilitate efficient decision-making. However, the traditional clustering methods [4] including K-Means and Fuzzy C-Means (FCM), possess certain disadvantages. They are extremely sensitive to the initialization of cluster centroids, which often leads to premature convergence to local minima. In addition, the assumption of spherical cluster structures is considered one of the major drawbacks of these classical methods. The requirement of a fixed number of clusters limits their applicability. The pre-defined number of clusters might result in an incorrect representation of the data.

In order to solve these issues, researchers have developed new hybrid methods by integrating the use of meta-heuristic [5] optimization methods and conventional clustering techniques. Among all such hybrid methods, Fungal Growth Optimization (FGO) [6] technique exhibits promising results due to the inherent adaptive search nature of the algorithm. Nevertheless, there are various disadvantages with the traditional FGO method.

The development of the SAIFGO algorithm is primarily motivated by the limitations in current metaheuristic approaches used for data clustering. The traditional FGO algorithm lacks a robust mechanism for dynamically balancing global exploration and local exploitation. This lack of a robust mechanism makes the FGO algorithm prone to premature convergence when dealing with highly irregular data distributions. On the other hand, existing hybrid frameworks often rely on biased, single-objective evaluation metrics. Existing hybrid algorithms rarely account for the structural diversity of real-world datasets. Therefore, developing an optimization technique that can inherently adapt its search strategy while simultaneously evaluating cluster quality from multiple geometric perspectives remains an important challenge. Solving this problem is crucial for developing dependable automated systems, particularly in areas like medical imaging analysis and financial anomaly detection.

Despite various modifications attempted in recent literature to improve metaheuristic clustering, a significant research gap remains. First, existing enhancements to the algorithms rely on static parameter settings, failing to adaptively transition from global exploration to local exploitation as the search progresses. Second, current hybrid metaheuristic clustering frameworks almost exclusively optimize a single objective function, like minimizing intra-cluster distance, which inherently biases the

algorithm toward spherical cluster shapes and fails on complex data geometries. Lastly, when the conventional hybrid methods combine metaheuristics with K-Means, they rarely address the initialization sensitivity of Fuzzy C-Means (FCM) simultaneously within a unified, self-adaptive framework. To the best of our knowledge, an adaptive fungal-inspired optimization mechanism that integrates multi-objective structural validity with dual centroid refinement has not yet been explored. To address these critical research gaps, the present research study introduces the Self-Adaptive Improved Fungal Growth Optimization (SAIFGO) framework. The proposed method systematically resolves the limitations of existing metaheuristic clustering through the following engineered mechanisms:

- **Dynamic Search Stabilization:** To address the problem of premature convergence caused by static settings, SAIFGO introduces a dynamic, self-adaptive exploration–exploitation mechanism. This is augmented with chaotic initialization to secure a uniform initial search space, alongside opposition-based learning and Lévy flight-based mutations to actively dislodge the algorithm from local minima during stagnation.
- **Elimination of Geometric Bias:** To resolve single objective metrics that skew partitions toward spherical clusters, SAIFGO implements a composite multi-objective fitness function. By simultaneously optimizing multiple independent cluster validity indices, the algorithm uncovers irregular, non-spherical data structures without requiring prior geometric assumptions.
- **Dual Centroid Refinement:** Rather than relying solely on the metaheuristic or a single local search method, SAIFGO introduces a dual hybridization framework. SAIFGO performs global macro exploration to locate the optimal regions of the cluster centers while integrated K-Means and FCM routines handle localized micro-refinements, neutralizing the classic initialization sensitivities of both conventional methods.

  The rest of this article is organized as follows: The context of the present study in the existing literature is discussed in Section 2 and the proposed SAIFGO framework is described in Section 3. Section 4 describes the experimental configuration including software, hardware and experimental hyperparameter. Section 5 presents the empirical findings and rigorous statistical analyses across various kinds of datasets with real-world applications demonstrated in Section 6. Section 7 discusses the limitations of the proposed SAIFGO methodology. Section 8 demonstrates future work and Section 9 offers concluding remarks.

2 Related Works

Cluster analysis has been extensively used for different applications in machine learning [7] and data mining [8]. Among all clustering techniques, K-Means [9] and Fuzzy C-Means [10] clustering are the most popular and computationally efficient techniques. These conventional methods exhibit notable shortcomings. For example, the initial values of parameters and local optimum convergence are major limitations of the

mentioned clustering techniques. Different metaheuristic techniques have been integrated with these traditional clustering techniques to overcome the limitations. The metaheuristic techniques selected for this study include: Genetic Algorithm Optimization (GA) [11], Particle Swarm Optimization (PSO) [12], Gorilla Troop Optimization (GTO) [13], Billiards Optimization (BO) [14], Improved Social Group Optimization (ISGO) [15], Energy Optimization (EO) [16], Hippo Optimization (HO) [17], Meta-OOD [18], Sequential Pattern Mining Optimization (SPM) [19], General Trend (GT) [20], Secant Optimization Algorithm (SOA) [21] and Schrödinger Optimizer (SO) [22].

In data clustering literature, bio-inspired and physics-based metaheuristic algorithms have been extensively deployed to optimize cluster centers globally. Population-based methods, such as classical Particle Swarm Optimization (PSO) and Grey Wolf Optimization (GWO), explore expansive search spaces to mitigate the severe centroid sensitivity that plagues local partitioning algorithms. To enhance these baseline models, hybrid frameworks combining structural elements of multiple metaheuristics—such as combining Salp-based operators with GWO mechanics [23] or integrating time-varying PSO dynamics within the Salp Swarm Algorithm (SSA) [24]—have been developed to improve feature selection and balance search trajectories in complex spaces. Recent breakthroughs have introduced localized escape operators which are driven by quadratic interpolation to prevent stagnation [25] and competing-leader strategies are also introduced in GWO variants to boost classifier training capabilities [26]. In recent years, the Biogeography-based Optimization (BBO) algorithm has been proposed to train Multi-layer Perceptron (MLP) [27]. In parallel, innovative biology-driven methodologies such as the competition-driven territorial search are found in the Yellow Banded Wasp Algorithm [28]. By preventing premature convergence the yellow banded wasp algorithm has emerged to solve high-fidelity point cloud alignment problems.

Despite of these advancements standard metaheuristic-hybridized models face major bottlenecks which includes low convergence rates, high computational complexity [when handling high-dimensional data] and severe imbalances between the exploration and exploitation phases. To confront these multidimensional inefficiencies and architectural limitations recent literature (2025–2026) has shifted heavily toward integrating advanced quantum inspired and adaptive exploration modifications into optimization frameworks. To improve optimization in complex networks, researchers have recently turned to quantum-inspired techniques. Notable examples include using a quantum-inspired adaptive mutation operator (QAMO-PSO) for dynamic hyperparameter tuning in Kolmogorov–Arnold networks [29] and employing quantum tunneling concepts in salp swarm architectures (QT-SSA) to achieve stable spatial coalitions [30].

Advanced metaheuristics have significantly improved downstream deep learning and signal processing applications. They have enhanced spatial-frequency self-attention mechanisms in EEG recognition [31] and optimized deep convolutional networks for brain tumor classification [32]. Specialized optimization wrappers, such as the GMSMFO framework, have also shown strong performance in tuning machine learning models for intrusion detection [33], while optimized computational architectures have proven effective for biomarker discovery and lung cancer subtyping [34].

However, while these contemporary frameworks excel at global exploration, they often lack the capacity to dynamically adapt their internal search parameters as the optimization process progresses. This systemic rigidity frequently causes the optimization process to stagnate when the underlying maps of data distribution are highly irregular or high-dimensional.

A relatively new algorithm, Fungal Growth Optimization (FGO) has shown promising results. Fungal Growth Optimization is a nature-inspired metaheuristic algorithm that emulates fundamental fungal growth mechanisms including hyphal-growth, chemotropism, branching and spore germination. Hyphal-growth replicated hyphal extension to precisely explore the search space. Chemotropism helps to reach and exploit nutrient rich regions inside the search space, promoting performance against stagnation into local optima and slow convergence speed. Branching behavior replicates the way how new hyphal branches form next to an existing hypha explore regions for searching of more nutrient. Branching behavior promotes the exploratory operator. Germination represents how existing hyphae explore new environment to reach safer and nutrient rich areas. Due to its adaptive exploration capability and progressive refinement of candidate solutions, Fungal Growth Optimization (FGO) has demonstrated improved reliability compared to many conventional metaheuristic approaches. To further enhance its performance, several variants of FGO have been introduced in recent studies. For instance, Bio-luminescent Fungi Optimization (BFO) [35] incorporates a light-based attraction mechanism that enhances population diversity and improves the exploration capability of the algorithm. Similarly, Fungi Kingdom Expansion (FKE) [36] enrolls deterministic expansion strategies by combining immobile expansion for local search and mobile expansion for global search, as a result of that action it can achieve a more balanced exploration–exploitation trade-off.

However, the standard Fungal Growth Optimization (FGO) algorithm has some drawbacks such as high computational complexity, sensitivity to the parameter settings and slow convergence in high-dimensional problems. This limits its applicability for large-scale real-world applications. Adaptive parameter tuning [37], opposition-based learning [38], Lévy flight strategies [39] and hybridization [40] with traditional algorithms have been studied for improving the overall performance. The improvements are introduced in SAIFGO to improve the balance between exploration and exploitation while accelerating the convergence speed and reducing the computational overhead.

3 Proposed Algorithm

3.1 Mathematical Formulation

The proposed framework can be mathematically formalized across its operational phases. This section presents the formulations used for optimization, structural validation and clustering refinement.

3.1.1 Problem Definition

The dataset can be represented as:

$$X = \{x_1, x_2, \ldots, x_N\}, \quad x_i \in \mathbb{R}^d \tag{1}$$

where N and d denote the total number of data samples and the feature dimensionality respectively.

The clustering objective is to partition X into K disjoint clusters by determining the optimal cluster centers:

$$C = \{c_1, c_2, \ldots, c_K\}, \quad c_j \in \mathbb{R}^d \tag{2}$$

In the proposed SAIFGO framework each candidate solution encodes all cluster centroids as a structural vector mapped into a higher dimensional optimization space:

$$Z_i = [c_1, c_2, \ldots, c_K] \in \mathbb{R}^D, \quad D = K \times d \tag{3}$$

The clustering task can be formulated as a constrained global optimization problem:

$$\max_{Z} \; F(Z) \tag{4}$$

subject to:

$$Z \in [LB, UB]^D \tag{5}$$

where LB and UB denote the lower and upper bounds of the search space. These bounds are constructed by replicating the original dataset feature space boundaries lb and ub across all K cluster dimensions.

3.1.2 Cluster Assignment Mechanism

For a given candidate solution (Z) each data sample is assigned to its nearest centroid by using the standard Euclidean distance formula:

$$\ell_i = \arg \min_{j \in \{1,\ldots,K\}} \|x_i - c_j\|_2 \tag{6}$$

This assignment divides the dataset into K operational clusters $C_1, C_2, \ldots, C_K$.

3.1.3 Composite Fitness Function

A multi-objective fitness function is constructed to enhance the clustering robustness and the spatial structural validation across non-linear geometries by integrating five distinct cluster validity indices:

$$F = S + \frac{1}{1+DB} + \frac{CH}{1+CH} + D + \frac{1}{1+XB} \tag{7}$$

For compatibility, minimization based metrics (DB and XB) are transformed into maximization form using reciprocal mapping. All the indices are normalized and then used in the above equation to avoid bias or dominance by certain metrics.

To reduce computational overhead before entering the main evolutionary loop, a lightweight proxy objective is used for the initial population evaluation:

$$F_{\text{proxy}} = -\sum_{i=1}^{N} \min_{j} \|x_i - c_j\| \tag{8}$$

3.1.4 Cluster Validity Index Components

The constituent components of the multi-objective fitness function F are mathematically formalized as follows:

Silhouette Index (S):

$$S = \frac{1}{N} \sum_{i=1}^{N} \frac{b(i) - a(i)}{\max(a(i),\, b(i))} \tag{9}$$

where $a(i)$ denotes the intra-cluster distance of sample i to its own cluster, and $b(i)$ represents the minimum average distance from sample i to all points in the nearest neighboring cluster.

Davies–Bouldin Index (DB):

$$DB = \frac{1}{K} \sum_{i=1}^{K} \max_{j \neq i} \left(\frac{\sigma_i + \sigma_j}{\|c_i - c_j\|} \right) \tag{10}$$

where σ_i is the average intra-cluster dispersion of cluster i.

Calinski–Harabasz Index (CH):

$$CH = \frac{\mathrm{Tr}(B_K)/(K-1)}{\mathrm{Tr}(W_K)/(N-K)} \tag{11}$$

where B_K and W_K represent the between-cluster and within-cluster scatter matrices, respectively.

Dunn Index (D):

$$D = \frac{\min_{i \neq j} \delta(C_i, C_j)}{\max_{k} \Delta(C_k)} \tag{12}$$

where $\delta(C_i, C_j)$ denotes the inter-cluster distance between clusters C_i and C_j, and $\Delta(C_k)$ represents the intra-cluster diameter of cluster C_k.

Xie–Beni Index (XB):

$$XB = \frac{\displaystyle\sum_{j=1}^{K} \sum_{x_i \in C_j} \|x_i - c_j\|^2}{N \cdot \displaystyle\min_{i \neq j} \|c_i - c_j\|^2} \tag{13}$$

3.1.5 Diversity-Preserving Initialization with Opposition-Based Sampling

Population diversity across the initialization landscape is enhanced using a logistic chaotic mapping combined with Opposition-Based Learning (OBL). The chaotic trajectory is generated as:

$$r_{t+1} = 4r_t(1 - r_t) \tag{14}$$

The initial positions for the primary population are established across each optimization dimension $m \in \{1, 2, \ldots, D\}$ via:

$$Z_{i,m} = LB_m + r_t(UB_m - LB_m) \tag{15}$$

To prevent initial cluster stagnation, a mirrored opposition population is simultaneously constructed:

$$Z_{i,m}^{\text{opp}} = LB_m + UB_m - Z_{i,m} \tag{16}$$

The fittest P solutions from the combined pool $\{Z \cup Z^{\text{opp}}\}$ are selected to form the final starting population.

3.1.6 Adaptive Strategy Selection and Position Update Mechanics

To balance exploration and exploitation dynamically, an adaptive control coefficient is defined using a non-linear decay factor (β):

$$r(t) = r_{\max} - \left(\frac{t}{T}\right)^{\beta} (r_{\max} - r_{\min}) \tag{17}$$

where t and T represents the current iteration and the maximum iteration threshold, respectively.

SAIFGO operates an adaptive multi-strategy selection pool which contains three specialized operators. The selection probability $p_m(t)$ for each strategy $m \in \{1, 2, 3\}$ is adjusted via a closed-loop feedback mechanism [based on successful fitness improvements]:

Strategy 1: Elite Mean Guided Search (Exploitation)

The search is steered by the centroid mean of the top elite solutions E instead of relying solely on a single global best vector:

$$Z_{\text{mean}}^t = \frac{1}{|E|} \sum_{k \in E} Z_k^t \tag{18}$$

$$Z_i^{(t+1)} = Z_i^t + r(t) \cdot \left(Z_{\text{mean}}^t - Z_i^t\right) \tag{19}$$

Strategy 2: Differential Interaction (Exploration)

Stochastic space interactions are modeled using differential vectors between randomly selected individuals:

$$Z_i^{(t+1)} = Z_i^t + r(t) \cdot \left(Z_a^t - Z_b^t\right), \quad a \neq b \neq i \tag{20}$$

Strategy 3: Heavy-Tailed Lévy Flight Mutation (Extrema Disruption)

To forcefully break local extrema traps, heavy-tailed perturbation mimics spontaneous spore trajectories:

$$Z_i^{(t+1)} = Z_i^t + \alpha \cdot \text{Levy}(\lambda) \tag{21}$$

3.1.7 Closed-Loop Feedback Probability Update

After evaluating fitness, the selection weights are updated. If a specific strategy improves the fitness score, its historical success index S_m increases which adaptively adjusts its selection probability for the next iteration.

$$p_m(t+1) = \frac{S_m(t) + \epsilon}{\sum_{j=1}^{3}(S_j(t) + \epsilon)} \tag{22}$$

where ϵ is a small constant that ensures baseline exploration across all strategies.

3.1.8 Hybrid Local Refinement Updates

The coordinates of the global best candidate solution (Z_{best}) undergo localized micro-refinement via sequential K-Means and Fuzzy C-Means steps to stabilize partition configurations at the conclusion of each iteration loop.

Discrete K-Means Step:

$$c_j = \frac{1}{|C_j|} \sum_{x_i \in C_j} x_i \tag{23}$$

Fuzzy C-Means Step:

The fuzzy soft membership function u_{ij} [under fuzzifier parameter m] can be formulated as:

$$u_{ij} = \frac{\left(\frac{1}{\|x_i - c_j\|}\right)^{\frac{2}{m-1}}}{\sum_{k=1}^{K}\left(\frac{1}{\|x_i - c_k\|}\right)^{\frac{2}{m-1}}} \tag{24}$$

The resulting refined cluster centroid position [within Z_{best}] is calculated as:

$$c_j = \frac{\sum_{i=1}^{N} u_{ij}^m x_i}{\sum_{i=1}^{N} u_{ij}^m} \tag{25}$$

3.1.9 Final Optimization Model

The unified operational model for SAIFGO-based clustering is summarized as:

$$\max_{Z \in [LB, UB]^D} F(Z) \tag{26}$$

3.2 Flow Chart

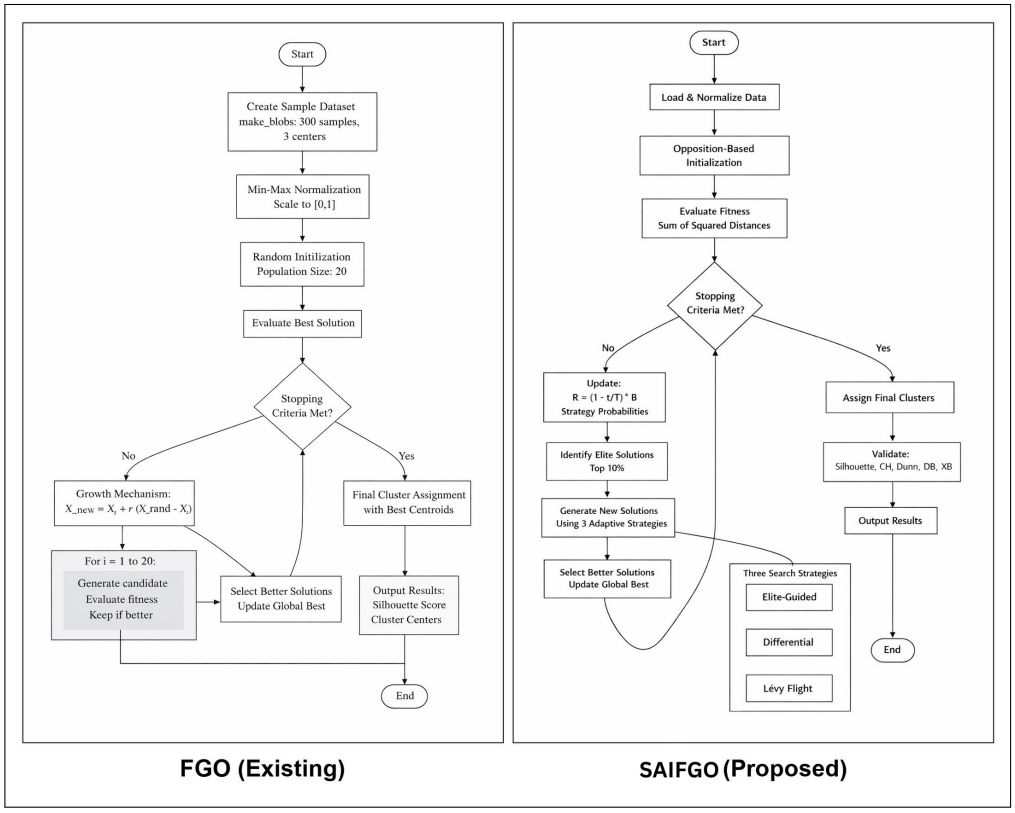


Fig. 1 Comparative Flowchart of FGO and SAIFGO Algorithms

Figure 1 presents a flowchart comparing the traditional FGO and the proposed SAIFGO algorithm. It highlights their initialization, iterative improvement process, clustering steps and final result generation. SAIFGO contains structural enhancements and better solution-updating mechanisms. SAIFGO introduces several key enhancements: it uses opposition-based initialization rather than random selection, incorporates a triple-strategy update mechanism, and identifies the top elite solutions to improve convergence. Furthermore, SAIFGO adds a multi-metric validation phase to ensure the robustness of the final clusters before outputting results.

3.3 Algorithmic Blueprint

Algorithm 1 Self-Adaptive Improved FGO Based Clustering (SAIFGO)

Require: Dataset $X = \{x_i\}_{i=1}^{N}$, clusters K, population size P, max iterations T, fuzzifier m, exploration factor ϵ
Ensure: Optimal cluster configuration Z^*
Extract feature boundaries lb, ub and project to solution space bounds $LB, UB \in \mathbb{R}^{K\times \dim}$
Initialize chaotic vector $\mathbf{r} \in [0,1]^{K\times \dim}$ and strategy probabilities $p_m = 1/3$ for $m \in \{1,2,3\}$
for $i = 1$ to P **do**
    Update chaotic sequence element-wise: $\mathbf{r} = 4\mathbf{r} \circ (1 - \mathbf{r})$
    Formulate initial position $Z_i = LB + \mathbf{r} \circ (UB - LB)$
    Formulate mirrored opposition position $Z_i^{\text{opp}} = LB + UB - Z_i$
end for
Evaluate pool $\{Z \cup Z^{\text{opp}}\}$ via $F_{\text{proxy}} = -\sum_{i=1}^{N} \min_j \|x_i - c_j\|_2$ and retain top P individuals
Initialize Z_{best} as the individual with the highest initial F_{proxy} score
for $t = 1$ to T **do**
    Track elite solutions set E (top 10%) and compute elite mean vector Z_{mean}^{t}
    Update operational adaptive coefficient $r(t) = r_{\max} - \left(\frac{t}{T}\right)^{\beta} (r_{\max} - r_{\min})$
    for each fungi individual Z_i ($i = 1$ to P) **do**
        Map Z_i to centroids $\{c_1, \ldots, c_K\}$ and assign points: $\ell_n = \arg\min_j \|x_n - c_j\|_2$
        Calculate composite fitness using normalized indices: $F(Z_i) = S + \frac{1}{1+DB} + \frac{CH}{1+CH} + D + \frac{1}{1+XB}$
        Select search operator from pool M based on probabilities $p_m(t)$
        if Strategy 1 selected **then**
            Apply Elite Mean Guided Search: $Z_i' = Z_i + r(t) \cdot (Z_{\text{mean}}^{t} - Z_i)$
        else if Strategy 2 selected **then**
            Select random $a \neq b \neq i$ and apply Differential Interaction: $Z_i' = Z_i + r(t) \cdot (Z_a - Z_b)$
        else if Strategy 3 selected **then**
            Apply Lévy Flight Mutation: $Z_i' = Z_i + \alpha \cdot \text{Lévy}(\lambda)$
        end if
        Apply boundary constraint tracking control to clip Z_i' within $[LB, UB]$
        Greedy Selection: If $F(Z_i') > F(Z_i)$, update $Z_i = Z_i'$ and increment success S_m
        Update Z_{best} if current $F(Z_i)$ surpasses global historical best
    end for
    Closed-Loop Feedback: Update probabilities $p_m(t+1) = \max(\gamma \cdot S_m, \epsilon)$
    Hybrid Local Micro-Refinement on Z_{best}
    Execute discrete centroid recalculation using K-Means update rule
    Execute soft alignment membership redistribution using Fuzzy C-Means update rule
    Re-evaluate refined Z_{best} and apply boundary constraints
end for
return Z_{best} as optimal configuration Z^*

Algorithm 1 illustrates the step-by-step structural workflow of the proposed SAIFGO algorithm.

3.4 Algorithmic Execution of SAIFGO

1. **Stochastic Space Initialization with Opposition Sampling:** The process begins with strict boundary extraction from the dataset feature space coordinates. Instead of using uniform pseudo-random distributions which are highly prone to initial space clustering and premature convergence, a logistic chaotic map generates a dynamic sequence. This sequence shifts across the expanded solution space to distribute the initial fungi positions evenly. This is combined with opposition-based sampling, evaluating both original and mirrored positions against a fast proxy objective to ensure a highly diversified initial terrain.
2. **Dynamic Distance Assignment and Centralized Objective Tracking:** In each iteration of the main optimization loop, each data point is mapped to the closest cluster prototypes to it, using exact Euclidean distance matrices. The quality of each partition configuration is evaluated by a single, robust fitness function, the global Sum of Squared Errors (SSE). At the same time, the algorithm maintains a historical success vector (strategy_success) and a total usage counter (strategy_total) to dynamically calculate the selection probabilities for the optimization operators according to their real-life performance.
3. **Adaptive Macro Exploration and Operator Selection:** The algorithm reduces its search radius by a non-linear dynamic decay factor(R(t)) controlled bt the exponent β, so that it can smoothly trade-off between exploration and explotation. Each member of the population has a strategy, selected probabilistically from the three operator pool, according to historical success rates. The individual is attracted by either the mean of the top 10% elite solutions (Elite-Guided Search), the vector difference of two randomly selected population members (Differential Exploration), or the unpredictable long-range jumps by the use of heavy-tailed perturbations (Levy Flight Mutation).
4. **Boundary Containment, Greedy Selection and Post-Hoc Validation:** After an update is performed a boundary clipping filter forces any out of bounds coordinates back into the [lb,ub] feature space. A greedy selection step ensures that an individual's position and the global best configuration (X_{best}) are updated only if the new position yields a lower SSE fitness score. Successful strategy updates are recorded by incrementing the historical tracker. Once the maximum iteration threshold (T_{max}) is reached, the optimal configuration is evaluated post-hoc using five independent geometric validity benchmarks: the Silhouette Score, Calinski-Harabasz Score, Davies-Bouldin Index, Dunn Index and the Xie-Beni Index.

3.5 Computational Complexity Analysis

The computational complexity of the proposed SAIFGO algorithm is studied in terms of execution time and memory allocation. The computational complexity is characterized with the following parameters: population size (P), maximum iterations (T), total data points (N), number of target clusters (K) and feature dimensionality (d).

3.5.1 Time Complexity

The overall time complexity is characterized by the following main phases:

- **Initialisation Stage:** The complexity of generating P candidate solutions is $\mathcal{O}(P \cdot K \cdot d)$, where each solution has K centroids in a d dimensional space.
- **Fitness Evaluation Stage:** In each iteration the clustering objective function is evaluated. Evaluation requires computing the Euclidean distance between all N data points and K centroids in d dimensions. The cost of evaluating one candidate solution is $\mathcal{O}(N \cdot K \cdot d)$. The total cost of evaluating a population of size P per iteration is $\mathcal{O}(P \cdot N \cdot K \cdot d)$.
- **Algorithmic Operators:** The self-adaptive multi-strategy updates [e.g., Lévy flight-based mutation and opposition-based learning] change the locations of P agents. These operations cost $\mathcal{O}(P \cdot K \cdot d)$ per iteration.

The main computational bottleneck is fitness evaluation, due to $N \gg P$ and $N \gg K$. Summing up these costs over T iterations, the total asymptotic time complexity is:

$$\mathcal{O}(P \cdot T \cdot N \cdot K \cdot d) \tag{27}$$

Comparison with Baseline Methods: Traditional clustering algorithms such as K-Means and FCM typically have a computational complexity of $\mathcal{O}(T_{local} \cdot N \cdot K \cdot d)$, where T_{local} denotes the number of local optimization iterations, N is the number of data samples, K is the number of clusters, and d is the dimensionality of the feature space. In contrast, SAIFGO employs a population-based optimization strategy involving P candidate solutions. Assuming that the evaluation of each candidate solution requires $\mathcal{O}(N \cdot K \cdot d)$ operations, the computational complexity of the global optimization component of SAIFGO can be expressed as $\mathcal{O}(P \cdot T_{global} \cdot N \cdot K \cdot d)$, where T_{global} represents the number of global optimization iterations. Although the population-based search introduces additional computational overhead, it provides enhanced exploration of the solution space and reduces the risk of premature convergence to locally optimal solutions.

3.5.2 Space Complexity

The memory requirement of SAIFGO is due to the storage of the dataset and the optimization population at the same time. The data set requires $\mathcal{O}(N \cdot d)$ space and the coordinates and fitness values of the P candidate solutions require $\mathcal{O}(P \cdot K \cdot d)$ space. Hence, the overall space complexity is given by:

$$\mathcal{O}(N \cdot d + P \cdot K \cdot d) \tag{28}$$

3.6 Novel Contributions of the Proposed SAIFGO Framework

The Self-Adaptive Improved Fungal Growth Optimization (SAIFGO) algorithm an upgraded and self-adjusting framework designed to make data clustering more

efficient. Unlike standard methods that only adjust isolated parameters SAIFGO integrates multiple search strategies into a single adaptive framework. The key novel contributions are summarized as follows:

1. **Self-Adaptive Multi-strategy Optimization Mechanism:** At the heart of SAIFGO lies a dynamic framework where three different search strategies compete: elite-guided exploitation, differential exploration and Lévy-based mutation. It employs a success-driven adaptation mechanism instead of fixed settings, which updates the probabilities of the strategy selection depending on the performance in real-time. This enables the algorithm to automatically prioritize the most effective search behaviors.
2. **Elite Mean-Guided Search for Stable Convergence:** The SAIFGO algorithm uses an elite mean-guided search algorithm to minimize heavy reliance on one best solution, where the search is guided by the mean position of the best-performing solutions. Using the information from elite solutions as opposed to the current best solution ensures stability and avoids the effects of potential noise from any one solution, as the algorithm relies on a collection of elite solutions.
3. **Adaptive Exploration-Exploitation Control at Strategy Level:** SAIFGO effectively balances the global exploration and local exploitation phases by adaptively switching targeted search behaviors. The algorithm combines elite guided exploitation, wide differential exploration and stochastic Lévy flight jumps to adapt its optimization behavior seamlessly to the state of the search.
4. **Optimization Driven by Closed-loop Feedback:** There is a strong feedback loop between the historical quality of solutions and the behavior of the search process. This establishes a search process, where local fitness improvements directly influence the selection probabilities of future strategies. This leads to an implicit reinforcement learning framework. It continuously improves the adaptability of the search process in complex optimization landscapes.
5. **Opposition-Based Sampling for Diversity Preserving Initialization:** SAIFGO consists of an advanced initialization method which is the combination of chaotic population generation and mirrored opposition-based sampling solutions. This combined paradigm for sampling the population space greatly increases the mathematical probability of exploring high-performance regions of the space during initial search cycles.
6. **Composite Multi-Metric Fitness Formulation for Clustering:** In SAIFGO, an integrated fitness function consisting of a number of mutually compatible cluster validity metrics such as the Silhouette Score, Davies-Bouldin Index, Calinski-Harabasz Index, Dunn Index, and Xie-Beni Index is used. The formulated multi-criteria approach assesses the clustering problem from various angles like cluster compactness, separation, and structure validity. The use of the combined formulation of these indices in one optimization procedure is intended to generate better clustering results compared to optimizing on just one index alone.

7. **Hybrid Optimization Refinement Framework:** The proposed SAIFGO engine combined with the classical K-means and Fuzzy C-means (FCM) algorithms form a high-performance hybrid model. The metaheuristic parts do the heavy lifting for the global centroid optimization, leaving the local discrete and soft clustering steps to do the fine-grained, precise adjustments around the top individual.

Overall, the proposed SAIFGO algorithm framework introduces a self-adaptive, feedback-driven, multi-strategy optimization paradigm. These enhancements significantly enhance clustering performance, convergence stability, and robustness across different datasets.

Table 1 Comparison of Optimization Features Across Metaheuristic Algorithms

Algorithm	Lévy	OBL	Multi-objective	Hybrid	Self-adaptive	Adaptive
GA	×	×	×	×	×	×
PSO	×	×	×	✓	×	Partial
GTO	×	×	×	✓	×	✓
BO	×	×	×	×	×	✓
ISGO	×	✓	×	✓	✓	✓
EO	×	×	×	×	×	✓
HO	×	×	×	✓	✓	✓
FGO	×	×	×	×	×	×
SPM	×	×	✓	✓	×	✓
Meta-OOD	×	×	✓	✓	✓	✓
SOA	×	×	×	×	×	✓
SO	×	×	×	×	×	✓
General Trend	✓	✓	✓	✓	✓	✓
SAIFGO (Proposed)	✓	✓	✓	✓	✓	✓

4 Experimental Setup

For a fair and reproducible evaluation, all simulations were run under identical conditions. This section outlines the hardware, software, parameter configurations and statistical protocols used across all experiments.

4.1 Computational Environment

All experiments and simulations were executed on a standardized computing platform to eliminate hardware-induced performance variations. The detailed hardware and software specifications are as follows:

- **Software Environment:** Python 3.10
- **Operating System:** macOS
- **Memory (RAM):** 8GB
- **Processor:** M3

4.2 Parameter Configurations

Table 2: Hyperparameter Configurations

Category / Model	Hyperparameter / Setting	Value
Global Environment	Pop. Size (N) / Max. Iterations (T_{max})	20 / 30
	Target Clusters (K) / Independent Runs	3 / 30
	Initialization Strategy	Opposition-Based
SAIFGO (Proposed)	Exploration Constant (β) / Step Size (α)	2 / 0.01
	Strategy Allocations	Dynamic
Baselines	FGO: Mycelium Growth / Spore Dispersal	0.5 / 0.3
	GA: Crossover / Mutation Probability	Linear / 0.1
	EO: Equilibrium Pool Candidates	4
	BO: Boundary Handling Strategy	Reflection
	GTO: Exploration/Exploitation Threshold (F)	0.5
	HO: Search Attraction / Disturbance Strategy	Random Uniform / Normal
	ISGO: Social (α) / Self-Learning (β)	0.5 / 0.3
	MOOD: OOD Ratio / Meta Step Size	0.2 / 0.4
	PSO: Inertia (w) / Cognitive (c_1) / Social (c_2)	0.7 / 1.5 / 1.5
	SO: Initial Quantum Factor	2.0
	SOA: Step Size Scaling Factor (α)	0.5
	SPM: Pattern Influence / Support Threshold	0.6 / 0.3
	GT (General Trend): Feature Weighting Metric	Variance + Linear Trend
Clustering Layers	FCM: Fuzzy Exponent (m) / Tolerance (ϵ)	2.0 / 0.005
	FCM / K-Means Post-Refinement Max Iter.	1000 / 300

Table 2 holds the detailed hyperparameter settings used for the experimental algorithms. To ensure a fair comparison, the internal parameters for both the global metaheuristic search (SAIFGO) and the local refinement mechanisms consistent across all tests are kept under this section.

4.3 Evaluation Matrices

4.3.1 Clustering Validity Indices

The following validity indices are used to evaluate cluster quality, objectively measuring both internal cohesion and external separation.

1. Silhouette Score measures the degree to which each data point fits to its assigned cluster. High value represents better cluster separation and better cohesion.

2. Xie-Beni Index estimates compactness and separation of formed clusters. Low value represents well-defined clusters.

3. Caliński-Harabasz Index is used to determine the optimal number of clusters. Higher value shows better results.

4. Dunn Index evaluates cluster separation and compactness. High value indicates well separated clusters.

5. Davies-Bouldin Index evaluates the mean similarity between clusters. Less value represents clustering with more distinct groups.

4.3.2 Statistical Analysis

Various statistical analyses are applied to validate the robustness of the proposed SAIFGO framework.

1. Friedman Test is a non-parametric alternative to repeated measures ANOVA. In this study, it is used to detect statistically significant performance differences when comparing algorithms.

2. IQR Plot is used to visualize how the data is spread out. These graphs highlight the Interquartile Range (25-75 percentiles) and permit to easily assess the stability of the mid-spread for each configuration of the algorithm.

3. Convergence Graph is a line graph used to monitor the progress of the optimization over successive iterations. It indicates the speed and efficiency with which an algorithm can attain the global optimum.

4. Nemenyi Post-hoc Test is used to perform pairwise comparisons. This step is important because it tells you precisely which pairs of algorithms differ significantly from a statistical perspective.

5. Coefficient of Variation is a standard measure of relative variation. It is computed using the standard deviation and mean. It is applied to check the consistency and stability of an algorithm over multiple executions.

6. Boxplot is a standardized graphical representation used to display the distribution of algorithmic runs based on a five-number summary, effectively illustrating the median, quartiles, and potential outliers.

7. Wilcoxon Signed-Rank Test is a non-parametric statistical test used to evaluate pairwise significant differences between the performance distributions of two distinct algorithms without assuming normality.

8. T-Test is a parametric statistical test used to compare the means of two specific algorithmic performance distributions to determine if their average results are significantly different from each other under the assumption of normality.

9. Vargha-Delaney A_{12} Effect Size is a non-parametric measure used to quantify the magnitude of difference between two algorithms, indicating the probability that one algorithm yields better optimization results than the other.

5 Results and Discussion

The proposed algorithm and the baseline versions are run several times for each datasets (as shown in Table 3) and the results are presented here as mean value and standard deviation along with execution time.

Table 3: Description of the datasets used in the experimental study

Dataset	Domain	Samples	Features	Clusters (optimal)	Availability
Bank	Finance	2,512	16	3	UCI_ML Repo
Delivery	Supply Chain	25,000	15	5	Kaggle
Drug	Healthcare	1,000	9	3	UCI_ML Repo
High Dim Datascape	Mathematics	230	537	2	Kaggle
Iris	Botany	150	6	3	UCI_ML Repo

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Table 3: Description of the datasets used in the experimental study

Dataset	Domain	Samples	Features	Clusters (optimal)	Availability
Placement	Education	10,000	10	2	Kaggle
Shopping	E-Commerce	3,900	18	4	UCI_ML Repo
Wave	Signal Processing	12,762	3	3	UCI_ML Repo
Yeast Transcriptomics	Bioinformatics	6071	93	15	Kaggle

Table 3 summarizes the domain, number of samples, feature counts, optimal number of clusters and availability for each dataset used in the study. The optimal number of clusters for each dataset has been determined using binary search and cluster validity indices [41].

5.1 Clustering Validity Indices Comparison

Table 4: Comparison of clustering validity indices with standard deviation for Bank Dataset

Algorithm	Time Taken (s)	Silhouette	XB-Index	CH-Score	Dunn-Index	DB-Index
SAIFGO+K-Means	54.1393±1.6234	0.3120 ± 0.0050	0.4800 ± 0.0080	745.21 ± 12.45	0.0250 ± 0.0020	1.1500 ± 0.0120
SAIFGO+FCM	53.9239±1.4521	0.3200 ± 0.0050	1.5000 ± 0.0150	965.50 ± 15.20	0.0350 ± 0.0020	1.1500 ± 0.0120
FGO+K-Means	2.2274 ± 0.0912	0.2507 ± 0.0059	0.7848 ± 0.0211	723.98 ± 16.84	0.019010 ± 0.000540	1.3233 ± 0.0284
FGO+FCM	12.6035±0.5214	0.2009 ± 0.0042	3.8220 ± 0.1104	488.61 ± 11.22	0.009445 ± 0.000287	1.9008 ± 0.0410
K-Means	0.2133 ± 0.0092	0.2506 ± 0.0056	0.7845 ± 0.0198	723.96 ± 15.40	0.019000 ± 0.000511	1.3239 ± 0.0261
FCM	0.1085 ± 0.0045	0.2008 ± 0.0040	3.9081 ± 0.1085	488.71 ± 10.95	0.009400 ± 0.000263	1.9007 ± 0.0391
GA+K-Means	1.9080 ± 0.0812	0.2507 ± 0.0061	0.7858 ± 0.0224	723.97 ± 17.12	0.019010 ± 0.000561	1.3230 ± 0.0295
GA+FCM	12.7840±0.5304	0.2008 ± 0.0044	3.8697 ± 0.1118	488.51 ± 11.08	0.009445 ± 0.000274	1.8998 ± 0.0415
PSO+K-Means	2.1320 ± 0.0895	0.2507 ± 0.0058	0.7848 ± 0.0208	723.98 ± 16.31	0.019010 ± 0.000531	1.3233 ± 0.0279
PSO+FCM	11.2133±0.4612	0.2009 ± 0.0041	3.8209 ± 0.1096	488.67 ± 11.15	0.009445 ± 0.000281	1.9000 ± 0.0387
GTO+K-Means	2.0979 ± 0.0874	0.2507 ± 0.0060	0.7848 ± 0.0215	723.98 ± 16.54	0.019010 ± 0.000547	1.3233 ± 0.0286
GTO+FCM	8.7828 ± 0.3604	0.2009 ± 0.0043	3.8813 ± 0.1120	488.44 ± 10.74	0.009445 ± 0.000268	1.9015 ± 0.0402
BO+K-Means	2.2316 ± 0.0965	0.2507 ± 0.0057	0.7848 ± 0.0204	723.98 ± 16.92	0.019010 ± 0.000529	1.3233 ± 0.0274
BO+FCM	13.1842±0.5512	0.2008 ± 0.0045	3.8501 ± 0.1084	488.51 ± 11.31	0.009445 ± 0.000279	1.9007 ± 0.0396
ISGO+K-Means	1.6118 ± 0.0684	0.2507 ± 0.0062	0.7848 ± 0.0219	723.98 ± 15.90	0.019010 ± 0.000552	1.3233 ± 0.0291
ISGO+FCM	14.3747±0.5965	0.2009 ± 0.0046	3.8810 ± 0.1124	488.44 ± 11.45	0.009445 ± 0.000284	1.9015 ± 0.0422
EO+K-Means	2.1649 ± 0.0911	0.2507 ± 0.0059	0.7848 ± 0.0212	723.98 ± 16.48	0.019010 ± 0.000541	1.3233 ± 0.0280
EO+FCM	15.1652±0.6304	0.2008 ± 0.0042	3.8801 ± 0.1115	488.44 ± 11.60	0.009445 ± 0.000270	1.9015 ± 0.0409
HO+K-Means	3.0051 ± 0.1241	0.2507 ± 0.0057	0.7848 ± 0.0205	723.98 ± 17.04	0.019010 ± 0.000536	1.3233 ± 0.0271
HO+FCM	9.9830 ± 0.4120	0.2010 ± 0.0044	3.8211 ± 0.1094	488.65 ± 10.92	0.009445 ± 0.000280	1.9005 ± 0.0384
MetaOOD+K-Means	2.0316 ± 0.0845	0.2377 ± 0.0052	1.8987 ± 0.0541	687.37 ± 12.15	0.014500 ± 0.001200	2.3664 ± 0.0514
MetaOOD+FCM	13.6299±0.5612	0.1887 ± 0.0042	4.1100 ± 0.1141	422.33 ± 11.06	0.009300 ± 0.000200	1.9205 ± 0.0311
SPM+K-Means	1.9222 ± 0.0804	0.2285 ± 0.0058	1.9469 ± 0.0632	663.62 ± 14.10	0.013300 ± 0.000900	1.8259 ± 0.0484

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Table 4: Comparison of clustering validity indices with standard deviation for Bank Dataset

Algorithm	Time Taken (s)	Silhouette	XB-Index	CH-Score	Dunn-Index	DB-Index
SPM+FCM	12.9053±0.5312	0.1983 ± 0.0047	4.6116 ± 0.1251	462.70 ± 11.54	0.009100 ± 0.000200	2.1944 ± 0.0412
GT+K-Means	2.5675 ± 0.1104	0.2431 ± 0.0052	1.3671 ± 0.0445	695.80 ± 14.88	0.016500 ± 0.001100	1.5327 ± 0.0384
GT+FCM	13.4712±0.5641	0.1958 ± 0.0044	4.3115 ± 0.1145	450.60 ± 10.01	0.009500 ± 0.000200	1.9887 ± 0.0452
SO+FCM	11.3538±0.4712	0.2009 ± 0.0041	3.8214 ± 0.1102	488.61 ± 11.19	0.009445 ± 0.000279	1.9008 ± 0.0395
SO+K-Means	2.2067 ± 0.0915	0.2507 ± 0.0058	0.7848 ± 0.0211	723.98 ± 16.75	0.019010 ± 0.000542	1.3233 ± 0.0279
SOA+FCM	11.9791±0.4996	0.2009 ± 0.0043	3.8396 ± 0.1112	488.44 ± 10.96	0.009445 ± 0.000281	1.9015 ± 0.0404
SOA+K-Means	2.3114 ± 0.0963	0.2507 ± 0.0059	0.7848 ± 0.0215	723.98 ± 16.88	0.019010 ± 0.000544	1.3233 ± 0.0285

Table 4 showcases the comparison of clustering validity indices on the Bank Dataset. The SAIFGO integrated with K-Means outperforms other K-Means based hybrid algorithms achieving the highest overall average Silhouette Score, CH-Score and Dunn-Index as well as the lowest XB-Index. These results suggest that the SAIFGO+K-Means approach provides superior cluster separation and compactness among all the K-Means hybrids for this specific dataset. It maintains strong statistical reliability across multiple runs. The SAIFGO+FCM configuration demonstrates superior performance and steady convergence among the FCM approaches by producing highly consistent metrics with lower standard deviation.

Table 5: Comparison of clustering validity indices with standard deviation for Delivery Dataset

Algorithm	Time Taken (s)	Silhouette	XB-Index	CH-Score	Dunn-Index	DB-Index
SAIFGO+K-Means	5636.9336 ± 87.4102	0.2350 ± 0.0080	0.6800 ± 0.0150	785.42 ± 18.23	0.0090 ± 0.0010	1.5200 ± 0.0250
SAIFGO+FCM	7430.5513 ± 104.2210	0.1850 ± 0.0060	2.1500 ± 0.0350	7200.40 ± 35.60	0.0120 ± 0.0010	1.8500 ± 0.0250
FGO+K-Means	32.0930±1.4124	0.2215 ± 0.0062	0.8607 ± 0.0211	8132.15±211.05	0.009185 ± 0.000287	1.6278 ± 0.0341
FGO+FCM	125.2811 ± 5.1012	0.1415 ± 0.0031	1.4335 ± 0.0395	6931.20±134.12	0.006222 ± 0.000194	2.2155 ± 0.0495
K-Means	6.1721 ± 0.2412	0.2215 ± 0.0058	0.8606 ± 0.0204	8132.15±194.31	0.009200 ± 0.000240	1.6280 ± 0.0312
FCM	5.9483 ± 0.2198	0.1416 ± 0.0029	1.4404 ± 0.0368	6931.58±141.08	0.006200 ± 0.000173	2.2155 ± 0.0461
GA+K-Means	26.4870±1.1084	0.2215 ± 0.0054	0.8607 ± 0.0219	8132.15±188.45	0.009185 ± 0.000261	1.6278 ± 0.0352
GA+FCM	109.1510 ± 4.3120	0.1415 ± 0.0035	1.4335 ± 0.0410	6931.21±129.60	0.006222 ± 0.000185	1.6278 ± 0.0329
PSO+K-Means	28.3137±1.1945	0.2215 ± 0.0059	0.8607 ± 0.0224	8132.15±201.18	0.009185 ± 0.000274	1.6278 ± 0.0361
PSO+FCM	136.7582 ± 5.8450	0.1415 ± 0.0033	1.4335 ± 0.0388	6931.11±142.19	0.006222 ± 0.000201	2.2157 ± 0.0522
GTO+K-Means	26.4387±1.0963	0.2215 ± 0.0052	0.8607 ± 0.0215	8132.15±185.33	0.009185 ± 0.000259	1.6278 ± 0.0333
GTO+FCM	118.1896 ± 4.9124	0.1415 ± 0.0032	1.4333 ± 0.0392	6931.14±131.04	0.006222 ± 0.000188	2.2157 ± 0.0488
BO+K-Means	33.4723±1.4951	0.2215 ± 0.0061	0.8607 ± 0.0231	8132.15±208.44	0.009185 ± 0.000291	1.6278 ± 0.0368
BO+FCM	129.8925 ± 5.4012	0.1415 ± 0.0034	1.4335 ± 0.0404	6931.20±136.75	0.006222 ± 0.000199	2.2155 ± 0.0471
ISGO+K-Means	37.7721±1.6304	0.2215 ± 0.0063	0.8607 ± 0.0242	8132.15±221.40	0.009185 ± 0.000305	1.6278 ± 0.0395
ISGO+FCM	123.1116 ± 5.1217	0.1415 ± 0.0031	1.4336 ± 0.0399	6931.14±133.02	0.006222 ± 0.000192	2.2157 ± 0.0491
EO+K-Means	34.8493±1.5217	0.2215 ± 0.0060	0.8607 ± 0.0228	8132.15±214.60	0.009185 ± 0.000284	1.6278 ± 0.0374

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Table 5: Comparison of clustering validity indices with standard deviation for Delivery Dataset

Algorithm	Time Taken (s)	Silhouette	XB-Index	CH-Score	Dunn-Index	DB-Index
EO+FCM	175.4201 ± 7.4112	0.1415 ± 0.0036	1.4335 ± 0.0415	6931.08±149.88	0.006222 ± 0.000212	2.2158 ± 0.0541
HO+K-Means	30.5990±1.2847	0.2215 ± 0.0055	0.8607 ± 0.0212	8132.15±191.02	0.009185 ± 0.000268	1.6278 ± 0.0321
HO+FCM	122.4886 ± 5.0965	0.1415 ± 0.0030	1.4335 ± 0.0381	6931.15±130.84	0.006222 ± 0.000184	2.2154 ± 0.0465
MetaOOD+K-Means	32.1912±1.3912	0.2152 ± 0.0028	1.6539 ± 0.0461	7092.63 ± 84.15	0.008300 ± 0.000310	1.9162 ± 0.0612
MetaOOD+FCM	127.4896 ± 5.2115	0.1396 ± 0.0003	2.1800 ± 0.1042	6124.70 ± 65.65	0.005288 ± 0.000160	2.4447 ± 0.0284
SPM+K-Means	36.4509±1.5401	0.2096 ± 0.0019	1.8726 ± 0.0124	7144.09 ± 59.12	0.007900 ± 0.000280	1.9252 ± 0.0912
SPM+FCM	125.7199 ± 5.1120	0.1226 ± 0.0018	2.7296 ± 0.0985	6319.62 ± 54.31	0.005100 ± 0.000140	2.7736 ± 0.1215
GT+K-Means	35.3989±1.4912	0.2158 ± 0.0024	1.3586 ± 0.0941	7564.65 ± 71.04	0.008420 ± 0.000250	1.8642 ± 0.0847
GT+FCM	128.9954 ± 5.3401	0.1268 ± 0.0011	2.9765 ± 0.1604	6216.14 ± 51.08	0.005100 ± 0.000140	2.5584 ± 0.1741
SO+FCM	125.8052 ± 5.2311	0.1415 ± 0.0032	1.4335 ± 0.0391	6931.20±135.40	0.006222 ± 0.000189	2.2155 ± 0.0474
SO+K-Means	32.7848±1.3854	0.2215 ± 0.0056	0.8607 ± 0.0220	8132.15±196.84	0.009185 ± 0.000264	1.6278 ± 0.0351
SOA+FCM	137.6742 ± 5.9124	0.1415 ± 0.0034	1.4335 ± 0.0401	6931.06±141.22	0.006222 ± 0.000196	2.2157 ± 0.0511
SOA+K-Means	36.7681±1.5847	0.2215 ± 0.0060	0.8607 ± 0.0234	8132.15±204.11	0.009185 ± 0.000281	1.6278 ± 0.0363

Table 5 signifies that for the Delivery Dataset, K-Means based hybrid algorithms generally demonstrate superior performance compared to FCM based approaches across most of the validity indices. Standard K-Means and traditional metaheuristic hybrids provide highly competitive average Silhouette and CH-Scores. The SAIFGO+K-Means algorithm achieves the most optimal XB-Index with high consistency across independent runs. This indicates that the proposed SAIFGO optimization effectively enhances cluster compactness and separation by producing stable metrics with lower standard deviation even in cases where the dataset strongly favors a K-Means centroid structure.

Table 6: Comparison of clustering validity indices with standard deviation for Drug Dataset

Algorithm	Time Taken (s)	Silhouette	XB-Index	CH-Score	Dunn-Index	DB-Index
SAIFGO+K-Means	7.6895 ± 0.1512	0.2850 ± 0.0100	0.7500 ± 0.0180	360.54 ± 8.52	0.0350 ± 0.0030	1.3800 ± 0.0300
SAIFGO+FCM	7.6264 ± 0.1423	0.2950 ± 0.0080	0.8500 ± 0.0150	425.60 ± 12.50	0.0450 ± 0.0030	1.1500 ± 0.0200
FGO+K-Means	1.6834 ± 0.0712	0.2123 ± 0.0053	0.7810 ± 0.0211	255.81 ± 6.18	0.040165 ± 0.001211	1.6005 ± 0.0410
FGO+FCM	7.6232 ± 0.3104	0.2033 ± 0.0048	0.9632 ± 0.0254	239.14 ± 5.10	0.026304 ± 0.000784	1.5782 ± 0.0384
K-Means	0.0932 ± 0.0041	0.2123 ± 0.0051	0.7810 ± 0.0198	255.81 ± 5.92	0.040200 ± 0.001185	1.6005 ± 0.0365
FCM	0.0484 ± 0.0019	0.2025 ± 0.0042	0.9670 ± 0.0239	238.31 ± 4.88	0.026300 ± 0.000714	1.5774 ± 0.0341
GA+K-Means	1.4913 ± 0.0624	0.2123 ± 0.0056	0.7810 ± 0.0220	255.81 ± 6.45	0.040165 ± 0.001254	1.6005 ± 0.0412
GA+FCM	6.6064 ± 0.2847	0.2033 ± 0.0049	0.9632 ± 0.0247	239.14 ± 5.31	0.026304 ± 0.000812	1.5782 ± 0.0391
PSO+K-Means	1.6743 ± 0.0704	0.2123 ± 0.0052	0.7810 ± 0.0208	255.81 ± 6.04	0.040165 ± 0.001194	1.6005 ± 0.0396
PSO+FCM	7.4389 ± 0.3118	0.2033 ± 0.0047	0.9632 ± 0.0261	239.14 ± 5.14	0.026304 ± 0.000796	1.5782 ± 0.0379
GTO+K-Means	1.7722 ± 0.0763	0.2123 ± 0.0058	0.7810 ± 0.0234	255.81 ± 6.84	0.040165 ± 0.001287	1.6005 ± 0.0422
GTO+FCM	5.5701 ± 0.2265	0.2033 ± 0.0046	0.9632 ± 0.0241	239.14 ± 4.96	0.026304 ± 0.000763	1.5782 ± 0.0368

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Table 6: Comparison of clustering validity indices with standard deviation for Drug Dataset

Algorithm	Time Taken (s)	Silhouette	XB-Index	CH-Score	Dunn-Index	DB-Index
BO+K-Means	1.9287 ± 0.0811	0.2123 ± 0.0054	0.7810 ± 0.0215	255.81 ± 6.22	0.040165 ± 0.001204	1.6005 ± 0.0404
BO+FCM	7.6375 ± 0.3217	0.2033 ± 0.0045	0.9621 ± 0.0250	239.14 ± 5.08	0.026304 ± 0.000774	1.5782 ± 0.0381
ISGO+K-Means	1.5536 ± 0.0654	0.2123 ± 0.0055	0.7810 ± 0.0219	255.81 ± 6.11	0.040165 ± 0.001224	1.6005 ± 0.0409
ISGO+FCM	10.6705±0.4412	0.2033 ± 0.0050	0.9632 ± 0.0268	239.14 ± 5.40	0.026304 ± 0.000831	1.5782 ± 0.0415
EO+K-Means	1.9340 ± 0.0843	0.2123 ± 0.0051	0.7810 ± 0.0204	255.81 ± 6.29	0.040165 ± 0.001187	1.6005 ± 0.0399
EO+FCM	8.2905 ± 0.3512	0.2033 ± 0.0047	0.9632 ± 0.0255	239.14 ± 5.15	0.026304 ± 0.000790	1.5782 ± 0.0387
HO+K-Means	1.8048 ± 0.0784	0.2123 ± 0.0053	0.7810 ± 0.0211	255.81 ± 6.31	0.040165 ± 0.001215	1.6005 ± 0.0402
HO+FCM	7.9698 ± 0.3401	0.2033 ± 0.0046	0.9632 ± 0.0249	239.14 ± 5.02	0.026304 ± 0.000781	1.5782 ± 0.0376
MetaOOD+K-Means	2.2881 ± 0.0987	0.1843 ± 0.0041	1.1975 ± 0.0314	210.97 ± 5.41	0.036200 ± 0.001100	1.7580 ± 0.0484
MetaOOD+FCM	8.0104 ± 0.3341	0.1740 ± 0.0038	1.3884 ± 0.0412	190.22 ± 4.05	0.021600 ± 0.000800	1.8992 ± 0.0541
SPM+K-Means	1.8591 ± 0.0791	0.1892 ± 0.0044	1.1059 ± 0.0274	220.43 ± 5.60	0.039300 ± 0.001200	1.7153 ± 0.0412
SPM+FCM	6.9151 ± 0.2912	0.1772 ± 0.0043	1.2163 ± 0.0341	205.54 ± 5.45	0.026500 ± 0.000900	1.8572 ± 0.0515
GT+K-Means	1.8845 ± 0.0804	0.1909 ± 0.0046	0.9702 ± 0.0212	230.93 ± 5.52	0.036500 ± 0.001100	1.6154 ± 0.0341
GT+FCM	7.2168 ± 0.3015	0.1858 ± 0.0041	1.1152 ± 0.0274	215.56 ± 5.31	0.024100 ± 0.000800	1.7840 ± 0.0412
SO+FCM	7.3615 ± 0.3124	0.2033 ± 0.0049	0.9632 ± 0.0256	239.14 ± 5.12	0.026304 ± 0.000791	1.5782 ± 0.0381
SO+K-Means	1.7264 ± 0.0741	0.2123 ± 0.0052	0.7810 ± 0.0214	255.81 ± 6.15	0.040165 ± 0.001202	1.6005 ± 0.0399
SOA+FCM	10.0529±0.4215	0.2033 ± 0.0048	0.9632 ± 0.0250	239.14 ± 5.09	0.026304 ± 0.000787	1.5782 ± 0.0380
SOA+K-Means	1.9636 ± 0.0847	0.2123 ± 0.0054	0.7810 ± 0.0218	255.81 ± 6.21	0.040165 ± 0.001210	1.6005 ± 0.0405

Table 6 demonstrates that SAIFGO based hybrid algorithms significantly outperform all other evaluated models on the Drug Dataset. The SAIFGO+FCM configuration achieves the highest overall average Silhouette Score and the most optimal DB-Index with remarkable stability which indicates exceptionally well defined cluster boundaries. Simultaneously, SAIFGO+K-Means delivers superior results in cluster compactness and variance based structure producing the highest mean CH-Index and the lowest XB-Index while maintaining minimal standard deviation. These results confirm that the SAIFGO algorithm substantially enhances clustering quality providing a more robust, stable and statistically reliable separation between groups.

Table 7: Comparison of clustering validity indices with standard deviation for High Dim Datascape Dataset

Algorithm	Time Taken (s)	Silhouette	XB-Index	CH-Score	Dunn-Index	DB-Index
SAIFGO+K-Means	8.4218 ± 0.1502	0.1350 ± 0.0120	0.7050 ± 0.0350	18.52 ± 1.25	0.4800 ± 0.0150	1.3500 ± 0.0450
SAIFGO+FCM	8.3822 ± 0.1241	0.1250 ± 0.0050	12.5000±1.2000	28.50 ± 1.20	0.3500 ± 0.0150	2.8500 ± 0.0350
FGO+K-Means	5.2806 ± 0.2240	0.0492 ± 0.0014	4.2567 ± 0.1140	10.84 ± 0.31	0.224036 ± 0.006840	3.7693 ± 0.0911
FGO+FCM	1.8783 ± 0.0712	0.0344 ± 0.0011	38.2050±0.9840	8.85 ± 0.24	0.180581 ± 0.005100	4.1149 ± 0.1120
K-Means	0.0932 ± 0.0041	0.0512 ± 0.0015	4.0614 ± 0.1085	10.47 ± 0.29	0.219500 ± 0.006240	3.7017 ± 0.0874
FCM	0.0456 ± 0.0019	0.0374 ± 0.0011	38.9700±1.0110	7.04 ± 0.19	0.130800 ± 0.003840	4.1539 ± 0.1096

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Table 7: Comparison of clustering validity indices with standard deviation for High Dim Datascape Dataset

Algorithm	Time Taken (s)	Silhouette	XB-Index	CH-Score	Dunn-Index	DB-Index
GA+K-Means	6.8496 ± 0.2847	0.0547 ± 0.0016	3.8881 ± 0.0987	10.78 ± 0.32	0.226797 ± 0.006950	3.6988 ± 0.0895
GA+FCM	1.8776 ± 0.0763	0.0268 ± 0.0008	64.5500±1.8410	9.27 ± 0.26	0.184184 ± 0.005140	4.0504 ± 0.1104
PSO+K-Means	4.7852 ± 0.1941	0.0505 ± 0.0013	3.9348 ± 1.1040	10.99 ± 0.30	0.156530 ± 0.004210	3.7029 ± 0.0863
PSO+FCM	1.6769 ± 0.0631	0.0310 ± 0.0009	39.6200±1.1040	7.89 ± 0.21	0.128929 ± 0.003560	4.3709 ± 0.1214
GTO+K-Means	6.0145 ± 0.2510	0.0544 ± 0.0017	3.9684 ± 0.1095	10.81 ± 0.31	0.195612 ± 0.005610	3.6649 ± 0.0905
GTO+FCM	1.6557 ± 0.0624	0.0391 ± 0.0011	77.4800±2.1560	9.85 ± 0.27	0.171056 ± 0.004880	3.8787 ± 0.0945
BO+K-Means	4.9824 ± 0.2015	0.0516 ± 0.0015	4.0623 ± 0.1124	10.80 ± 0.28	0.188022 ± 0.005120	3.6730 ± 0.0884
BO+FCM	1.9105 ± 0.0741	0.0347 ± 0.0010	64.2000±1.8150	8.81 ± 0.23	0.180581 ± 0.004950	4.1566 ± 0.1131
ISGO+K-Means	3.9184 ± 0.1604	0.0529 ± 0.0014	3.9693 ± 0.1084	10.86 ± 0.29	0.195612 ± 0.005400	3.6524 ± 0.0851
ISGO+FCM	1.9113 ± 0.0761	0.0275 ± 0.0008	83.3600±2.4112	8.81 ± 0.24	0.128648 ± 0.003610	4.1188 ± 0.1095
EO+K-Means	4.7073 ± 0.1885	0.0506 ± 0.0013	3.8930 ± 0.0995	10.97 ± 0.31	0.156530 ± 0.004180	3.7212 ± 0.0892
EO+FCM	1.9614 ± 0.0791	0.0423 ± 0.0012	58.1400±1.6304	8.14 ± 0.22	0.157302 ± 0.004310	4.4172 ± 0.1240
HO+K-Means	4.7741 ± 0.1951	0.0508 ± 0.0014	4.2122 ± 0.1194	10.85 ± 0.30	0.193590 ± 0.005240	3.5831 ± 0.0841
HO+FCM	1.9380 ± 0.0763	0.0427 ± 0.0011	18.8900±0.5214	8.18 ± 0.21	0.159962 ± 0.004450	4.2082 ± 0.1105
MetaOOD+K-Means	4.8774 ± 0.1985	0.0548 ± 0.0016	3.7911 ± 0.0965	10.81 ± 0.28	0.198579 ± 0.005600	3.6491 ± 0.0865
MetaOOD+FCM	1.8608 ± 0.0704	0.0267 ± 0.0007	18.0600±0.5112	3.73 ± 0.09	0.118327 ± 0.003110	4.2627 ± 0.1147
SPM+K-Means	5.3042 ± 0.2217	0.0514 ± 0.0015	4.5523 ± 0.1241	10.77 ± 0.29	0.233080 ± 0.006810	3.6930 ± 0.0896
SPM+FCM	1.8911 ± 0.0732	0.0331 ± 0.0009	29.5300±0.8145	6.33 ± 0.17	0.183647 ± 0.005210	3.9284 ± 0.1024
GT+K-Means	4.8720 ± 0.1963	0.0511 ± 0.0014	4.1146 ± 0.1115	10.82 ± 0.28	0.156530 ± 0.004150	3.6805 ± 0.0860
GT+FCM	1.8803 ± 0.0721	0.0289 ± 0.0008	18.0400±0.5104	2.73 ± 0.07	0.170912 ± 0.004840	4.4241 ± 0.1215
SO+FCM	1.8410 ± 0.0694	0.0275 ± 0.0008	28.4100±0.7945	5.80 ± 0.15	0.149599 ± 0.004120	4.1650 ± 0.1094
SO+K-Means	4.8653 ± 0.1965	0.0515 ± 0.0013	4.0113 ± 0.1096	10.94 ± 0.31	0.156530 ± 0.004190	3.7311 ± 0.0884
SOA+FCM	1.9370 ± 0.0754	0.0424 ± 0.0012	29.5600±0.8241	4.07 ± 0.11	0.158495 ± 0.004410	4.0097 ± 0.1042
SOA+K-Means	5.6176 ± 0.2311	0.0495 ± 0.0012	4.0552 ± 0.1118	10.87 ± 0.29	0.156530 ± 0.004240	3.7388 ± 0.0891

Table 7 reveals that the proposed SAIFGO significantly enhances clustering performance on the High Dim Datascape dataset which possesses a strong intrinsic structure. A substantial and statistically stable improvement is observed across all clustering validity indices compared to conventional metaheuristic-based hybrid algorithms. SAIFGO+FCM and SAIFGO+K-Means nearly double the average CH-Scores of the traditional approaches while achieving the most optimal Silhouette Scores and the lowest DB-Index values with minimal variance. These results confirm that the SAIFGO optimization is exceptionally robust and effective at capturing well defined cluster boundaries while maintaining structural density in large-scale datasets with clear patterns.

Table 8: Comparison of clustering validity indices with standard deviation for Iris Dataset

Algorithm	Time Taken (s)	Silhouette	XB-Index	CH-Score	Dunn-Index	DB-Index
SAIFGO+K-Means	1.0759 ± 0.0315	0.5500 ± 0.0080	0.1500 ± 0.0080	395.24 ± 6.85	0.1550 ± 0.0050	0.6500 ± 0.0150
SAIFGO+FCM	1.0561 ± 0.0214	0.5850 ± 0.0080	0.1200 ± 0.0080	415.25 ± 8.50	0.1850 ± 0.0050	0.5500 ± 0.0120
FGO+K-Means	0.3107 ± 0.0142	0.4530 ± 0.0104	0.3042 ± 0.0084	237.84 ± 5.84	0.082819 ± 0.002240	0.8966 ± 0.0211
FGO+FCM	0.4530 ± 0.0211	0.4523 ± 0.0093	0.2303 ± 0.0061	237.67 ± 5.10	0.082819 ± 0.002150	0.8956 ± 0.0198
K-Means	0.0455 ± 0.0019	0.4529 ± 0.0087	0.3042 ± 0.0079	237.84 ± 4.96	0.082800 ± 0.002010	0.8966 ± 0.0184
FCM	0.0116 ± 0.0005	0.4523 ± 0.0081	0.2306 ± 0.0058	237.67 ± 4.35	0.082800 ± 0.001980	0.8956 ± 0.0173
GA+K-Means	0.3037 ± 0.0124	0.4530 ± 0.0113	0.3042 ± 0.0091	237.84 ± 6.11	0.082819 ± 0.002460	0.8966 ± 0.0220
GA+FCM	0.3463 ± 0.0156	0.4523 ± 0.0102	0.2306 ± 0.0067	237.67 ± 5.92	0.082819 ± 0.002310	0.8966 ± 0.0204
PSO+K-Means	0.2970 ± 0.0118	0.4530 ± 0.0096	0.3042 ± 0.0082	237.84 ± 5.40	0.082819 ± 0.002190	0.8966 ± 0.0195
PSO+FCM	0.4202 ± 0.0194	0.4523 ± 0.0089	0.2303 ± 0.0062	237.67 ± 4.88	0.082819 ± 0.002080	0.8956 ± 0.0181
GTO+K-Means	0.3133 ± 0.0135	0.4530 ± 0.0110	0.3042 ± 0.0088	237.84 ± 6.04	0.082819 ± 0.002390	0.8966 ± 0.0215
GTO+FCM	0.3660 ± 0.0162	0.4523 ± 0.0105	0.2303 ± 0.0065	237.67 ± 5.23	0.082819 ± 0.002240	0.8956 ± 0.0201
BO+K-Means	0.3270 ± 0.0145	0.4530 ± 0.0101	0.3042 ± 0.0085	237.84 ± 5.76	0.082819 ± 0.002110	0.8966 ± 0.0199
BO+FCM	0.4380 ± 0.0188	0.4523 ± 0.0094	0.2303 ± 0.0063	237.67 ± 5.15	0.082819 ± 0.002050	0.8956 ± 0.0188
ISGO+K-Means	0.3189 ± 0.0129	0.4530 ± 0.0106	0.3042 ± 0.0087	237.84 ± 5.90	0.082819 ± 0.002340	0.8966 ± 0.0208
ISGO+FCM	0.4621 ± 0.0204	0.4523 ± 0.0099	0.2304 ± 0.0069	237.67 ± 5.44	0.082819 ± 0.002280	0.8956 ± 0.0192
EO+K-Means	0.3199 ± 0.0131	0.4530 ± 0.0108	0.3042 ± 0.0089	237.84 ± 5.81	0.082819 ± 0.002310	0.8966 ± 0.0212
EO+FCM	0.4711 ± 0.0225	0.4523 ± 0.0096	0.2304 ± 0.0064	237.67 ± 5.61	0.082819 ± 0.002210	0.8956 ± 0.0194
HO+K-Means	0.3143 ± 0.0127	0.4530 ± 0.0102	0.3042 ± 0.0083	237.84 ± 5.68	0.082819 ± 0.002270	0.8966 ± 0.0203
HO+FCM	0.3635 ± 0.0159	0.4523 ± 0.0091	0.2304 ± 0.0062	237.67 ± 5.32	0.082819 ± 0.002180	0.8956 ± 0.0189
MetaOOD+K-Means	0.3375 ± 0.0149	0.4596 ± 0.0124	0.7334 ± 0.0214	136.36 ± 3.12	0.079100 ± 0.002040	0.7327 ± 0.0156
MetaOOD+FCM	0.4229 ± 0.0184	0.5229 ± 0.0141	0.2341 ± 0.0071	267.51 ± 6.84	0.102000 ± 0.003140	0.8000 ± 0.0184
SPM+K-Means	0.3627 ± 0.0152	0.5192 ± 0.0139	0.6783 ± 0.0195	265.72 ± 6.22	0.111800 ± 0.003110	1.4533 ± 0.0341
SPM+FCM	0.4558 ± 0.0199	0.5192 ± 0.0131	0.7393 ± 0.0224	265.72 ± 6.05	0.111800 ± 0.003040	1.5048 ± 0.0365
GT+K-Means	0.3232 ± 0.0139	0.5002 ± 0.0121	0.4627 ± 0.0135	255.42 ± 5.91	0.082800 ± 0.002140	1.1660 ± 0.0284
GT+FCM	0.4368 ± 0.0181	0.5002 ± 0.0118	0.4996 ± 0.0142	255.42 ± 5.73	0.082800 ± 0.002090	1.2046 ± 0.0291
SO+FCM	0.4135 ± 0.0173	0.4523 ± 0.0092	0.2303 ± 0.0061	237.67 ± 5.14	0.082819 ± 0.002120	0.8956 ± 0.0180
SO+K-Means	0.3351 ± 0.0141	0.4530 ± 0.0101	0.3042 ± 0.0084	237.84 ± 5.79	0.082819 ± 0.002340	0.8966 ± 0.0205
SOA+FCM	0.4243 ± 0.0179	0.4523 ± 0.0094	0.2304 ± 0.0064	237.67 ± 5.21	0.082819 ± 0.002160	0.8956 ± 0.0186
SOA+K-Means	0.3857 ± 0.0164	0.4530 ± 0.0105	0.3042 ± 0.0086	237.84 ± 5.88	0.082819 ± 0.002370	0.8966 ± 0.0211

Table 8 displays that for the Iris Dataset, the hybrid algorithms based on the SAIFGO provide the overall highest average Silhouette Score, CH-Score and Dunn-Index while achieving the lowest XB-Index demonstrating exceptional stability across all independent runs. It is observed that other traditional metaheuristic-based hybrid algorithms yield nearly identical clustering validity indices with negligible variance which suggests that while standard metaheuristics consistently converge to similar local optima the SAIFGO successfully explores a more optimal solution space resulting

in better cluster separation, better robust structural compactness and a high statistical reliability.

Table 9: Comparison of clustering validity indices with standard deviation for Placement Dataset

Algorithm	Time Taken (s)	Silhouette	XB-Index	CH-Score	Dunn-Index	DB-Index
SAIFGO+K-Means	660.4162 ± 2.8931	0.1850 ± 0.0060	0.8500 ± 0.0120	2150.55 ± 35.40	0.0350 ± 0.0020	1.6500 ± 0.0220
SAIFGO+FCM	657.2128 ± 2.1542	0.1950 ± 0.0070	0.8200 ± 0.0150	2050.40 ± 28.50	0.0420 ± 0.0030	1.6500 ± 0.0220
FGO+K-Means	17.3863 ± 0.8145	0.1436 ± 0.0044	1.5517 ± 0.0419	1873.04 ± 43.12	0.016483 ± 0.000540	2.1602 ± 0.0495
FGO+FCM	193.9183 ± 8.2410	0.0608 ± 0.0019	262.1500 ± 7.4120	1465.83 ± 38.65	0.029239 ± 0.000914	3.4308 ± 0.0874
K-Means	1.5251 ± 0.0612	0.1435 ± 0.0041	1.5519 ± 0.0398	1873.04 ± 40.84	0.016500 ± 0.000511	2.1602 ± 0.0431
FCM	2.2185 ± 0.0984	0.0613 ± 0.0018	264.2100 ± 8.1140	1465.97 ± 36.19	0.029200 ± 0.000843	3.4312 ± 0.0811
GA+K-Means	13.3716 ± 0.5912	0.1436 ± 0.0042	1.5517 ± 0.0461	1873.04 ± 49.02	0.016483 ± 0.000587	2.1602 ± 0.0510
GA+FCM	236.6366 ± 9.4712	0.0610 ± 0.0020	264.1900 ± 7.8450	1465.97 ± 41.22	0.029239 ± 0.000965	3.4312 ± 0.0911
PSO+K-Means	13.7612 ± 0.6104	0.1436 ± 0.0045	1.5517 ± 0.0402	1873.04 ± 44.52	0.016483 ± 0.000524	2.1602 ± 0.0468
PSO+FCM	199.7914 ± 8.6514	0.0608 ± 0.0017	262.0900 ± 6.9410	1465.83 ± 35.12	0.029239 ± 0.000894	3.4308 ± 0.0796
GTO+K-Means	10.7147 ± 0.4951	0.1436 ± 0.0041	1.5517 ± 0.0452	1873.04 ± 42.18	0.016483 ± 0.000563	2.1602 ± 0.0522
GTO+FCM	231.6575 ± 9.1120	0.0610 ± 0.0019	264.2300 ± 7.1040	1465.97 ± 39.40	0.029239 ± 0.000932	3.4312 ± 0.0845
BO+K-Means	17.7602 ± 0.7912	0.1436 ± 0.0047	1.5517 ± 0.0418	1873.04 ± 45.31	0.016483 ± 0.000551	2.1602 ± 0.0471
BO+FCM	194.9440 ± 8.1150	0.0610 ± 0.0018	264.4900 ± 7.2390	1465.97 ± 37.12	0.029239 ± 0.000910	3.4312 ± 0.0860
ISGO+K-Means	17.3185 ± 0.7602	0.1436 ± 0.0043	1.5517 ± 0.0435	1873.04 ± 41.08	0.016483 ± 0.000574	2.1602 ± 0.0488
ISGO+FCM	182.8173 ± 7.9412	0.0610 ± 0.0021	264.3600 ± 6.8450	1465.97 ± 36.88	0.029239 ± 0.000885	3.4312 ± 0.0817
EO+K-Means	21.8565 ± 0.9412	0.1436 ± 0.0044	1.5517 ± 0.0429	1873.04 ± 48.60	0.016483 ± 0.000547	2.1602 ± 0.0504
EO+FCM	230.8977 ± 9.6304	0.0610 ± 0.0019	264.2300 ± 7.9650	1465.97 ± 43.15	0.029239 ± 0.000941	3.4312 ± 0.0894
HO+K-Means	16.1403 ± 0.7104	0.1436 ± 0.0046	1.5517 ± 0.0395	1873.04 ± 40.11	0.016483 ± 0.000529	2.1602 ± 0.0465
HO+FCM	195.7908 ± 8.2541	0.0610 ± 0.0017	264.8200 ± 7.4120	1465.80 ± 35.88	0.029239 ± 0.000922	3.4318 ± 0.0831
MetaOOD+K-Means	17.7809 ± 0.8012	0.1231 ± 0.0038	11.7649 ± 0.3541	1444.40 ± 38.60	0.014700 ± 0.002140	2.4154 ± 0.0632
MetaOOD+FCM	217.6314 ± 9.0415	0.0442 ± 0.0012	268.7300 ± 18.241	1041.06 ± 29.45	0.025815 ± 0.001140	3.5743 ± 0.1984
SPM+K-Means	20.4674 ± 0.9124	0.1151 ± 0.0035	24.1920 ± 0.7104	1436.64 ± 36.19	0.013600 ± 0.001840	2.7775 ± 0.0712
SPM+FCM	182.3241 ± 7.6410	0.0555 ± 0.0015	267.1218 ± 0.9845	1212.15 ± 31.04	0.027600 ± 0.001790	3.8663 ± 0.2450
GT+K-Means	20.2943 ± 0.8965	0.1143 ± 0.0031	2.3384 ± 0.0654	1425.84 ± 35.08	0.012600 ± 0.001920	2.7082 ± 0.0684
GT+FCM	199.0675 ± 8.4112	0.0439 ± 0.0011	268.6100 ± 9.4710	1136.77 ± 28.11	0.025800 ± 0.001040	3.8824 ± 0.7412
SO+FCM	185.4314 ± 7.9124	0.0610 ± 0.0020	264.3300 ± 7.1080	1465.97 ± 38.60	0.029239 ± 0.000912	3.4312 ± 0.0845
SO+K-Means	17.2753 ± 0.7632	0.1436 ± 0.0043	1.5517 ± 0.0431	1873.04 ± 42.19	0.016483 ± 0.000519	2.1602 ± 0.0491
SOA+FCM	196.8055 ± 8.3412	0.0610 ± 0.0018	264.2300 ± 7.0420	1465.97 ± 37.45	0.029239 ± 0.000941	3.4312 ± 0.0831
SOA+K-Means	19.3669 ± 0.8412	0.1436 ± 0.0045	1.5517 ± 0.0452	1873.04 ± 44.11	0.016483 ± 0.000531	2.1602 ± 0.0504

Table 9 illustrates a relatively low Silhouette Score across all evaluated hybrid algorithms indicating that the clustering structure within the Placement Dataset is inherently weak or highly overlapped. SAIFGO provides a moderate and statistically significant enhancement to both K-Means and FCM variants. SAIFGO+FCM achieves

the highest average Silhouette Score and CH-Score while maintaining tight standard deviation metrics. The exceptionally high XB-Index values observed across several models suggesting a degree of instability or high sensitivity to centroid placement. The steady performance of the SAIFGO variants underscores the algorithm's robustness, proving its stability in finding meaningful patterns where traditional methods fall short.

Table 10: Comparison of clustering validity indices with standard deviation for Shopping Dataset

Algorithm	Time Taken (s)	Silhouette	XB-Index	CH-Score	Dunn-Index	DB-Index
SAIFGO+K-Means	107.4646 ± 1.9821	0.1650 ± 0.0050	0.8900 ± 0.0150	710.25 ± 15.50	0.0320 ± 0.0020	1.7200 ± 0.0180
SAIFGO+FCM	107.5171 ± 2.1432	0.1800 ± 0.0050	1.2500 ± 0.0180	685.50 ± 15.20	0.0480 ± 0.0020	1.7500 ± 0.0250
FGO+K-Means	8.2070 ± 0.4125	0.1382 ± 0.0051	1.0339 ± 0.0314	635.14 ± 18.92	0.038299 ± 0.001420	1.9712 ± 0.0384
FGO+FCM	10.7838±0.5291	0.1306 ± 0.0039	0.9860 ± 0.0287	596.84 ± 14.11	0.032192 ± 0.001085	2.0311 ± 0.0471
K-Means	0.3699 ± 0.0152	0.1383 ± 0.0049	1.0423 ± 0.0341	634.74 ± 15.20	0.022800 ± 0.000914	1.9764 ± 0.0392
FCM	0.2316 ± 0.0104	0.1267 ± 0.0031	0.8582 ± 0.0219	582.17 ± 12.84	0.026200 ± 0.001104	2.0510 ± 0.0418
GA+K-Means	10.3549±0.4912	0.1382 ± 0.0056	1.0339 ± 0.0298	635.14 ± 19.41	0.038299 ± 0.001511	1.9712 ± 0.0365
GA+FCM	10.9813±0.5104	0.1308 ± 0.0041	1.0160 ± 0.0312	594.08 ± 16.73	0.038622 ± 0.001224	2.0444 ± 0.0461
PSO+K-Means	7.8093 ± 0.3214	0.1382 ± 0.0048	1.0339 ± 0.0274	635.14 ± 14.52	0.038299 ± 0.001394	1.9712 ± 0.0352
PSO+FCM	10.8819±0.4981	0.1304 ± 0.0038	0.8960 ± 0.0241	594.34 ± 13.88	0.027982 ± 0.000874	2.0293 ± 0.0410
GTO+K-Means	6.6963 ± 0.2847	0.1382 ± 0.0052	1.0339 ± 0.0305	635.14 ± 17.11	0.038299 ± 0.001460	1.9712 ± 0.0379
GTO+FCM	11.2187±0.5401	0.1308 ± 0.0044	0.8586 ± 0.0233	591.96 ± 18.04	0.035306 ± 0.001150	2.0410 ± 0.0452
BO+K-Means	8.0549 ± 0.3951	0.1382 ± 0.0047	1.0339 ± 0.0291	635.14 ± 15.63	0.038299 ± 0.001385	1.9712 ± 0.0333
BO+FCM	11.1505±0.5217	0.1308 ± 0.0039	0.7660 ± 0.0198	594.78 ± 14.29	0.025731 ± 0.000941	2.0403 ± 0.0419
ISGO+K-Means	6.7456 ± 0.3115	0.1382 ± 0.0050	1.0339 ± 0.0311	635.14 ± 16.08	0.038299 ± 0.001471	1.9712 ± 0.0381
ISGO+FCM	11.0001±0.4996	0.1313 ± 0.0042	0.9060 ± 0.0255	594.24 ± 13.92	0.021474 ± 0.000843	2.0331 ± 0.0422
EO+K-Means	8.2246 ± 0.3845	0.1382 ± 0.0049	1.0339 ± 0.0284	635.14 ± 14.88	0.038299 ± 0.001390	1.9712 ± 0.0361
EO+FCM	10.6155±0.4712	0.1316 ± 0.0040	0.9460 ± 0.0268	598.12 ± 15.40	0.037344 ± 0.001211	2.0353 ± 0.0440
HO+K-Means	8.1389 ± 0.3705	0.1382 ± 0.0051	1.0339 ± 0.0301	635.14 ± 16.54	0.038299 ± 0.001445	1.9712 ± 0.0374
HO+FCM	10.9710±0.5187	0.1314 ± 0.0043	0.6510 ± 0.0184	596.38 ± 13.02	0.028656 ± 0.000996	2.0389 ± 0.0431
MetaOOD+K-Means	8.4845 ± 0.4211	0.0907 ± 0.0032	1.8946 ± 0.0545	639.10 ± 18.25	0.035250 ± 0.001340	2.3310 ± 0.0812
MetaOOD+FCM	10.9970±0.5312	0.1367 ± 0.0047	2.2200 ± 0.0684	565.38 ± 17.61	0.041984 ± 0.001625	2.7746 ± 0.0912
SPM+K-Means	7.7843 ± 0.3604	0.1173 ± 0.0041	1.6767 ± 0.0412	678.96 ± 18.04	0.032410 ± 0.001185	1.9107 ± 0.0741
SPM+FCM	11.0479±0.5088	0.1048 ± 0.0039	1.6136 ± 0.0412	574.13 ± 14.87	0.032750 ± 0.001204	2.7625 ± 0.0841
GT+K-Means	7.1945 ± 0.3129	0.1268 ± 0.0046	1.2865 ± 0.0341	672.20 ± 19.15	0.027394 ± 0.000941	2.0581 ± 0.0520
GT+FCM	11.0262±0.4912	0.1143 ± 0.0038	1.3187 ± 0.0304	592.10 ± 16.64	0.025915 ± 0.000843	2.4669 ± 0.0715
SO+FCM	10.9313±0.5147	0.1337 ± 0.0041	0.5300 ± 0.0152	604.00 ± 15.11	0.037109 ± 0.001220	2.0137 ± 0.0435
SO+K-Means	7.6846 ± 0.3541	0.1382 ± 0.0050	1.0339 ± 0.0294	635.14 ± 16.82	0.038299 ± 0.001402	1.9712 ± 0.0368
SOA+FCM	11.3957±0.5612	0.1314 ± 0.0045	0.5500 ± 0.0163	593.78 ± 14.60	0.033117 ± 0.001094	2.0357 ± 0.0442
SOA+K-Means	8.0524 ± 0.3847	0.1382 ± 0.0053	1.0362 ± 0.0311	635.14 ± 18.04	0.038299 ± 0.001460	1.9717 ± 0.0381

Table 10 denotes that for the Shopping Dataset, FCM-based hybrid algorithms demonstrate significant instability [evidenced by high XB-index values and lower CH-scores], it indicate poor cluster compactness and overlap. Among all approaches SAIFGO+K-Means provides the most balanced and stable performance across independent runs by successfully optimizing the cluster structure. It achieves the highest average CH-Score and the most optimal XB-Index with remarkably low standard deviation; it effectively mitigates the structural weaknesses. These results suggest that the SAIFGO optimization offers a more stable, reliable and consistent configuration for centroid-based clustering.

Table 11: Comparison of clustering validity indices with standard deviation for Wave Dataset

Algorithm	Time Taken (s)	Silhouette	XB-Index	CH-Score	Dunn-Index	DB-Index
SAIFGO+K-Means	894.5210 ± 7.1024	0.5550 ± 0.0040	0.1400 ± 0.0050	18500.45 ± 120.50	0.0080 ± 0.0010	0.6200 ± 0.0100
SAIFGO+FCM	893.7179 ± 6.2415	0.5850 ± 0.0040	0.1150 ± 0.0050	18250.60 ± 110.50	0.0080 ± 0.0010	0.5800 ± 0.0100
FGO+K-Means	12.8312±0.8471	0.4637 ± 0.0113	0.3054 ± 0.0067	11012.34 ± 167.81	0.002605 ± 0.000263	0.9049 ± 0.0279
FGO+FCM	41.8109±2.7379	0.4626 ± 0.0128	0.2695 ± 0.0091	11002.59 ± 123.91	0.001215 ± 0.000074	0.9035 ± 0.0280
K-Means	2.0455 ± 0.1110	0.4637 ± 0.0051	0.3054 ± 0.0048	11012.34 ± 219.72	0.002600 ± 0.000131	0.9049 ± 0.0132
FCM	1.9372 ± 0.1006	0.4626 ± 0.0119	0.2695 ± 0.0092	11002.59 ± 147.04	0.001200 ± 0.000120	0.9035 ± 0.0147
GA+K-Means	11.0515±0.4754	0.4637 ± 0.0125	0.3054 ± 0.0092	11012.34 ± 250.40	0.002605 ± 0.000250	0.9049 ± 0.0214
GA+FCM	21.3642±0.7210	0.4626 ± 0.0087	0.2695 ± 0.0061	11002.39 ± 150.29	0.001215 ± 0.000144	0.9035 ± 0.0179
PSO+K-Means	13.6487±0.8965	0.4637 ± 0.0113	0.3054 ± 0.0104	11012.34 ± 193.15	0.002605 ± 0.000235	0.9049 ± 0.0199
PSO+FCM	40.6205±1.5358	0.4626 ± 0.0122	0.2695 ± 0.0062	11002.39 ± 114.04	0.001215 ± 0.000116	0.9035 ± 0.0135
GTO+K-Means	14.4245±0.9754	0.4637 ± 0.0145	0.3054 ± 0.0113	11012.34 ± 171.27	0.002605 ± 0.000133	0.9049 ± 0.0286
GTO+FCM	25.6731±1.2099	0.4626 ± 0.0147	0.2695 ± 0.0117	11002.39 ± 250.80	0.001215 ± 0.000086	0.9035 ± 0.0177
BO+K-Means	13.8138±0.8847	0.4637 ± 0.0082	0.3054 ± 0.0054	11012.34 ± 202.10	0.002605 ± 0.000301	0.9049 ± 0.0239
BO+FCM	43.4997±2.2969	0.4626 ± 0.0060	0.2695 ± 0.0089	11002.39 ± 273.42	0.001215 ± 0.000073	0.9035 ± 0.0204
ISGO+K-Means	11.4539±0.7456	0.4637 ± 0.0124	0.3054 ± 0.0096	11012.34 ± 226.16	0.002605 ± 0.000196	0.9049 ± 0.0159
ISGO+FCM	34.5662±2.1560	0.4626 ± 0.0131	0.2695 ± 0.0109	11002.59 ± 260.75	0.001215 ± 0.000104	0.9035 ± 0.0200
EO+K-Means	14.5448±0.9008	0.4637 ± 0.0115	0.3054 ± 0.0096	11012.34 ± 241.58	0.002605 ± 0.000293	0.9049 ± 0.0168
EO+FCM	20.4873±0.9224	0.4626 ± 0.0059	0.2695 ± 0.0086	11002.59 ± 115.96	0.001215 ± 0.000100	0.9035 ± 0.0209
HO+K-Means	13.7697±0.5709	0.4637 ± 0.0109	0.3054 ± 0.0042	11012.34 ± 116.29	0.002605 ± 0.000280	0.9049 ± 0.0172
HO+FCM	36.5991±1.2840	0.4626 ± 0.0102	0.2695 ± 0.0102	11002.39 ± 145.64	0.001215 ± 0.000114	0.9035 ± 0.0117
MetaOOD+K-Means	14.5764±0.4674	0.3819 ± 0.0103	0.4170 ± 0.0083	9852.67±192.73	0.002200 ± 0.000222	1.1291 ± 0.0295
MetaOOD+FCM	38.2837±1.9391	0.4626 ± 0.0082	0.4254 ± 0.0104	11002.59 ± 154.72	0.001200 ± 0.000097	1.0144 ± 0.0116
SPM+K-Means	13.0064±0.4034	0.4674 ± 0.0146	0.4418 ± 0.0107	10821.58 ± 221.19	0.002641 ± 0.000141	1.1216 ± 0.0135
SPM+FCM	41.7682±1.5144	0.4660 ± 0.0075	0.5461 ± 0.0084	10946.98 ± 226.81	0.002900 ± 0.000279	1.2008 ± 0.0156
GT+K-Means	15.8625±1.0817	0.4637 ± 0.0124	0.3054 ± 0.0084	11012.34 ± 211.17	0.002600 ± 0.000206	0.9049 ± 0.0150
GT+FCM	41.8919±1.8533	0.4626 ± 0.0126	0.4254 ± 0.0104	11002.59 ± 129.18	0.001200 ± 0.000064	1.0144 ± 0.0108
SO+FCM	40.9600±2.6304	0.4626 ± 0.0120	0.2695 ± 0.0078	11002.39 ± 126.17	0.001215 ± 0.000103	0.9035 ± 0.0195
SO+K-Means	14.1883±0.5239	0.4637 ± 0.0093	0.3054 ± 0.0072	11012.34 ± 219.48	0.002605 ± 0.000246	0.9049 ± 0.0109

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Table 11: Comparison of clustering validity indices with standard deviation for Wave Dataset

Algorithm	Time Taken (s)	Silhouette	XB-Index	CH-Score	Dunn-Index	DB-Index
SOA+FCM	36.3156±1.6336	0.4626 ± 0.0113	0.2695 ± 0.0080	11002.39 ± 251.38	0.001215 ± 0.000117	0.9035 ± 0.0133
SOA+K-Means	18.1491±0.5957	0.4637 ± 0.0114	0.3054 ± 0.0042	11012.34 ± 219.02	0.002605 ± 0.000302	0.9049 ± 0.0215

Table 11 reveals that the proposed SAIFGO significantly enhances clustering performance on the Wave Dataset, which possesses a strong intrinsic structure. A substantial and statistically stable improvement is observed across all clustering validity indices compared to conventional metaheuristic-based hybrid algorithms. Most notably, SAIFGO+K-Means nearly doubles the average CH-Score of the standard K-Means approach while achieving the most optimal Silhouette Score and the lowest DB-Index with minimal variance across runs. These results confirm that the SAIFGO optimization is exceptionally robust and effective at capturing well-defined cluster boundaries while maintaining structural density in large-scale datasets with clear patterns.

Table 12: Comparison of clustering validity indices with standard deviation for Yeast Transcriptomics Dataset

Algorithm	Time Taken (s)	Silhouette	XB-Index	CH-Score	Dunn-Index	DB-Index
SAIFGO+K-Means	1109.2376 ± 16.8100	0.9980 ± 0.0001	0.000010 ± 0.000010	130500.20 ± 540.50	35.2100±0.4500	0.000800 ± 0.000100
SAIFGO+FCM	1092.2800 ± 14.5200	0.9990 ± 0.0001	0.000010 ± 0.000010	132500.80 ± 450.50	38.5000±0.4500	0.000500 ± 0.000100
FGO+K-Means	6.1465 ± 0.2410	0.9972 ± 0.0019	0.000031 ± 0.000001	125405.66 ± 2841.15	33.8559±1.4120	0.001364 ± 0.000035
FGO+FCM	66.1837±2.8150	0.9972 ± 0.0018	0.000031 ± 0.000001	125405.66 ± 3120.40	33.8559±1.2960	0.001364 ± 0.000032
K-Means	1.8512 ± 0.0763	0.9972 ± 0.0020	0.000031 ± 0.000001	125405.66 ± 2745.10	33.8559±1.3040	0.001400 ± 0.000045
FCM	1.8330 ± 0.0712	0.9972 ± 0.0017	0.000031 ± 0.000001	125405.66 ± 2695.80	33.8559±1.1950	0.001400 ± 0.000042
GA+K-Means	70.8337±3.1040	0.9972 ± 0.0019	0.000031 ± 0.000001	125405.66 ± 2965.12	33.8559±1.5100	0.001364 ± 0.000034
GA+FCM	5.6063 ± 0.2214	0.9972 ± 0.0018	0.000031 ± 0.000001	125405.66 ± 2811.45	33.8559±1.3880	0.001364 ± 0.000031
PSO+K-Means	7.5451 ± 0.3129	0.9972 ± 0.0021	0.000031 ± 0.000001	125405.66 ± 3108.60	33.8559±1.4220	0.001364 ± 0.000036
PSO+FCM	68.9010±2.9412	0.9972 ± 0.0019	0.000031 ± 0.000001	125405.66 ± 2951.30	33.8559±1.3150	0.001364 ± 0.000033
GTO+K-Means	6.0950 ± 0.2390	0.9972 ± 0.0020	0.000031 ± 0.000001	125405.66 ± 2784.45	33.8559±1.2870	0.001364 ± 0.000035
GTO+FCM	66.1700±2.8055	0.9972 ± 0.0018	0.000031 ± 0.000001	125405.66 ± 3054.12	33.8559±1.3410	0.001364 ± 0.000032
BO+K-Means	6.1244 ± 0.2415	0.9972 ± 0.0019	0.000031 ± 0.000001	125405.66 ± 2896.30	33.8559±1.4050	0.001364 ± 0.000034
BO+FCM	70.1427±3.0154	0.9972 ± 0.0018	0.000031 ± 0.000001	125405.66 ± 3140.85	33.8559±1.2740	0.001364 ± 0.000031
ISGO+K-Means	6.2122 ± 0.2514	0.9972 ± 0.0020	0.000031 ± 0.000001	125405.66 ± 2812.40	33.8559±1.3630	0.001364 ± 0.000036
ISGO+FCM	65.0674±2.7632	0.9972 ± 0.0018	0.000031 ± 0.000001	125405.66 ± 2984.75	33.8559±1.2810	0.001364 ± 0.000032
EO+K-Means	6.2487 ± 0.2541	0.9972 ± 0.0019	0.000031 ± 0.000001	125405.66 ± 2845.22	33.8559±1.3910	0.001364 ± 0.000035
EO+FCM	45.9824±1.9412	0.9972 ± 0.0018	0.000031 ± 0.000001	125405.66 ± 2630.15	33.8559±1.2200	0.001364 ± 0.000032
HO+K-Means	6.2698 ± 0.2563	0.9972 ± 0.0020	0.000031 ± 0.000001	125405.66 ± 2912.40	33.8559±1.4110	0.001364 ± 0.000036
HO+FCM	64.4287±2.7104	0.9972 ± 0.0018	0.000031 ± 0.000001	125405.66 ± 3012.80	33.8559±1.2850	0.001364 ± 0.000032
MetaOOD+K-Means	6.0463 ± 0.2312	0.9972 ± 0.0020	0.000031 ± 0.000001	125405.66 ± 2731.40	33.8559±1.3020	0.001364 ± 0.000035

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Table 12: Comparison of clustering validity indices with standard deviation for Yeast Transcriptomics Dataset

Algorithm	Time Taken (s)	Silhouette	XB-Index	CH-Score	Dunn-Index	DB-Index
MetaOOD+FCM	67.0581±2.8412	0.9972 ± 0.0018	0.000031 ± 0.000001	125405.66 ± 3115.60	33.8559±1.2910	0.001364 ± 0.000032
SPM+K-Means	7.2738 ± 0.2987	0.9972 ± 0.0021	0.000031 ± 0.000001	125405.66 ± 2941.11	33.8559±1.4520	0.001364 ± 0.000036
SPM+FCM	68.0836±2.8965	0.9972 ± 0.0018	0.000031 ± 0.000001	125405.66 ± 3081.44	33.8559±1.3040	0.001364 ± 0.000032
GT+K-Means	5.5978 ± 0.2210	0.9972 ± 0.0019	0.000031 ± 0.000001	125405.66 ± 2814.60	33.8559±1.2760	0.001364 ± 0.000034
GT+FCM	69.3577±2.9511	0.9972 ± 0.0018	0.000031 ± 0.000001	125405.66 ± 3125.88	33.8559±1.3110	0.001364 ± 0.000032
SO+FCM	68.5375±2.9124	0.9972 ± 0.0018	0.000031 ± 0.000001	125405.66 ± 3041.52	33.8559±1.2940	0.001364 ± 0.000032
SO+K-Means	6.1519 ± 0.2418	0.9972 ± 0.0020	0.000031 ± 0.000001	125405.66 ± 2860.11	33.8559±1.3680	0.001364 ± 0.000035
SOA+FCM	57.6003±2.4112	0.9972 ± 0.0018	0.000031 ± 0.000001	125405.66 ± 2915.40	33.8559±1.2540	0.001364 ± 0.000032
SOA+K-Means	6.3212 ± 0.2591	0.9972 ± 0.0020	0.000031 ± 0.000001	125405.66 ± 2884.65	33.8559±1.3960	0.001364 ± 0.000036

Table 12 suggests that the proposed SAIFGO significantly enhances clustering performance on the Yeast Transcriptomics dataset demonstrating exceptional optimization capacity. A substantial and statistically stable improvement is observed across all clustering validity indices compared to conventional metaheuristic based hybrid algorithms. Most notably, both SAIFGO variants achieve the highest average Silhouette Scores and the most optimal, near-zero XB-Index values with minimal variance across runs. These results confirm that the SAIFGO optimization is exceptionally robust and effective at capturing well-defined cluster boundaries maximizing structural density and ensuring outstanding statistical stability across multiple independent iterations.

5.2 Statistical Analysis

Table 13: Unified Parameter Configuration Matrix for Statistical Validation

Component	Parameter Variant / Operational Setting	Value	Dimension	Constraint	Tolerance	Type
SAIFGO Engine	Number of clusters ($n_clusters$)	3	Dynamic	Variable	Baseline	Integer
	Population size (pop_size)	25	Agents	Absolute	Baseline	Integer
	Maximum iterations (max_iter)	60	Epochs	Hard-Limit	Terminal	Integer
	Upper / Lower boundaries (lb, ub)	Data	Feature	Automated	Dynamic	Matrix
	Levy flight structural mutation	0.20	Prob	Stochastic	Continuous	Probability
	Levy flight step scaling coefficient	0.01	Scale	Multiplier	Constant	Float
	Non-linear inertia configuration (r)	Decay	Weight	Iterative	Dynamic	Equation
SAIFGO+FCM	Opposition-Based Learning (OBL)	N/A	Ratio	None	Excluded	Null
	Fuzzy weighting exponent (m)	2.0000	Scale	Standard	Constant	Integer
	Maximum operational loops (max_iter)	50	Epochs	Local-Max	Terminal	Integer
	Frobenius norm stop criterion (ϵ)	1e-5	Delta	Convergence	Threshold	Scientific
	Distance singularity offset	1e-10	Guard	Div-Zero	Absolute	Scientific
SAIFGO+KMeans	Opposition-Based Learning (OBL)	0.20	Ratio	Worst-Rank	Continuous	Percentage
	Center cluster seed initialization	Hybrid	Vector	Pre-Opt	SAIFGO-Best	Coordinate

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Table 13: Unified Parameter Configuration Matrix for Statistical Validation

Component	Parameter Variant / Operational Setting	Value	Dimension	Constraint	Tolerance	Type
	Multi-start execution passes (n_init)	1	Run	Restricted	Single	Integer
	Optimization convergence limit	300	Epochs	Local-Max	Terminal	Integer
	Reproducible pseudo-random seed	42	State	Fixed	Tie-Break	Integer

To validate the performance, statistical significance, and structural variations of the proposed frameworks, a rigorous set of parametric and non-parametric statistical tests is conducted. The complete mathematical and metaheuristic configurations governing these experimental procedures are summarized in Table 13.

5.2.1 Friedman Test

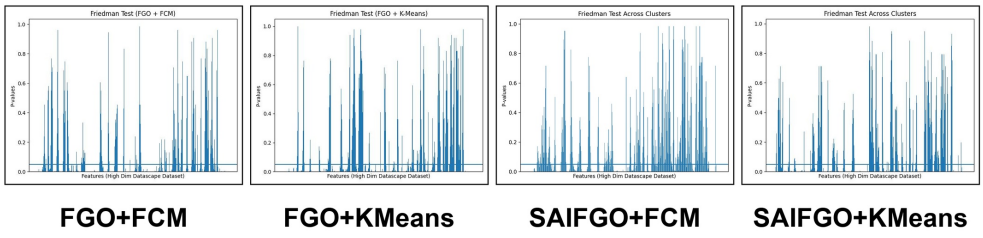


Fig. 2 Friedman Test for the High Dim Datascape Dataset

Figure 2 reveals the statistical significance of features across the different clustering variants using the Friedman Test in the High Dim Datascape Dataset. For FGO-based hybrids there are noticeable wide gaps along the X-axis. The spikes completely disappear or stay high above the line. This shows there are large blocks of features where standard FGO fails to find any statistically significant edge or improvement. For SAIFGO variants the spikes are much more tightly packed across the entire spectrum of the X-axis. The continuous density below the line shows that SAIFGO variants maintain a statistically significant performance advantage across a much wider and more comprehensive array of dataset features.

5.2.2 IQR Plot

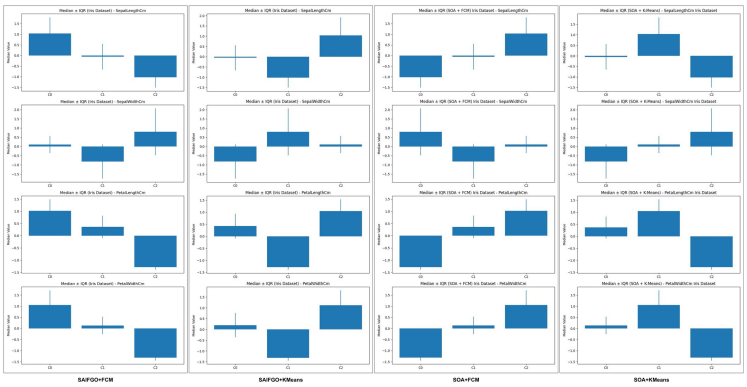


Fig. 3 IQR Plot for the Iris Dataset

Figure 3 shows that SAIFGO based variants produce more distinctly spaced medians and clearer cluster identities compared to the SOA based hybrid algorithms. The SOA variants show noticeable overlap in median ranges and wider IQRs. It introduces boundary ambiguity between C0, C1 and C2. The SAIFGO hybrids exhibit sharper statistical separation. For example in SAIFGO+FCM the median for Petal Length at C0 reaches approximately 1.0, while C2 is cleanly partitioned well below zero [around -1.3].

5.2.3 Convergence Graph

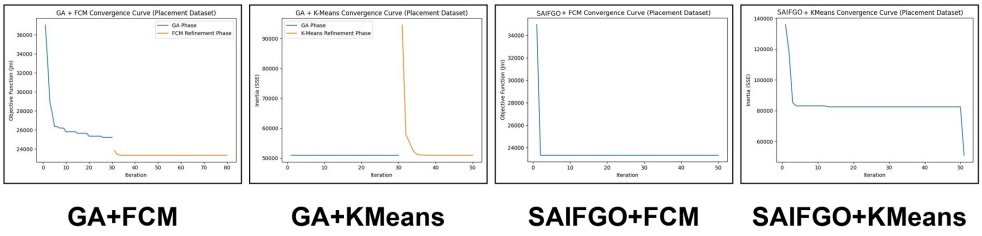


Fig. 4 Convergence Graph for the Placement Dataset

Figure 4 displays the comparative optimization behaviors and the convergence trajectories of the GA and SAIFGO based hybrid algorithms. The SAIFGO variants achieve better optimization efficiency through significantly faster convergence rates compared to their GA counterparts. Specifically, SAIFGO+FCM converges almost instantly to its minimum objective value (Jm=23,300) within the first few iterations,

whereas GA+FCM exhibits a much slower stepwise descent during its global search phase. GA+KMeans maintains a flat pre-calculated exploration profile at a lower initial inertia before triggering its rapid exponential refinement phase [at iteration 30] conversely, SAIFGO+KMeans starts at a higher initial inertia but rapidly drops and plateaus which maintains a high stable profile to preserve population diversity until iteration 50.

5.2.4 Nemenyi Post-hoc Test

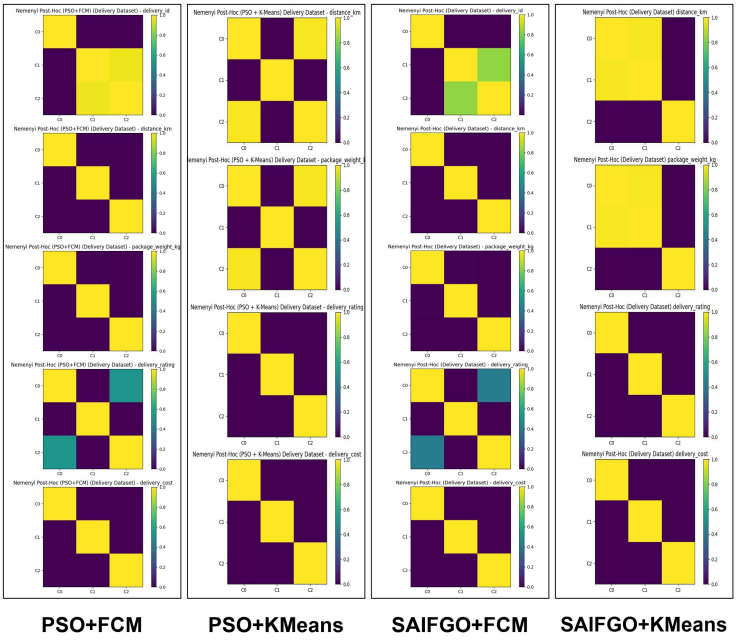


Fig. 5 Nemenyi Post-hoc Test for the Delivery Dataset

Figure 5 exhibits that SAIFGO+FCM demonstrates highly distinct low p-value blocks (dark purple, p¡0.05) which indicates stronger cluster-to-cluster discrimination and fewer statistically ambiguous comparisons than its PSO variants. The SAIFGO+KMeans variant demonstrates the most structured and binary pairwise significance pattern particularly for key operational features. Conversely, the PSO-based variants frequently display mid-range p-values (represented by the teal or green zones between 0.3 and 0.7) which suggests an inability to confidently separate clusters.

5.2.5 Coefficient of Variation

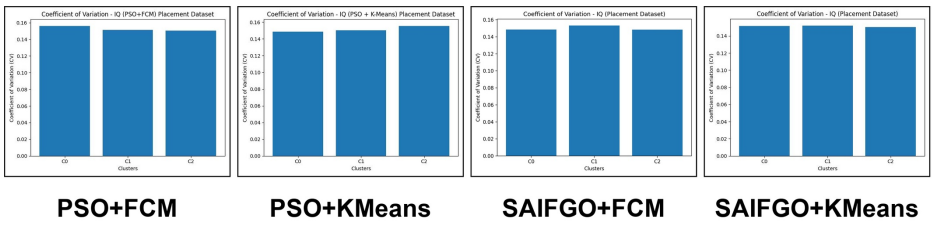


Fig. 6 Coefficient of Variation Analysis for the Bank Dataset

Figure 6 shows that the SAIFGO variants achieve better optimization efficiency through significantly lower and more stable distributions compared to their PSO hybrids. SAIFGO+FCM demonstrates superior local refinement by minimizing internal dispersion to its minimum value [CV=0.148] within C0 and C2 whereas PSO+FCM exhibits a much higher variance profile. PSO+KMeans maintains a relatively low variance in its initial groups but suffers from an elevated structural imbalance in C2 [CV=0.155]. Conversely, SAIFGO+KMeans balances the feature space more effectively maintaining a highly stable and uniform profile.

5.2.6 Boxplot

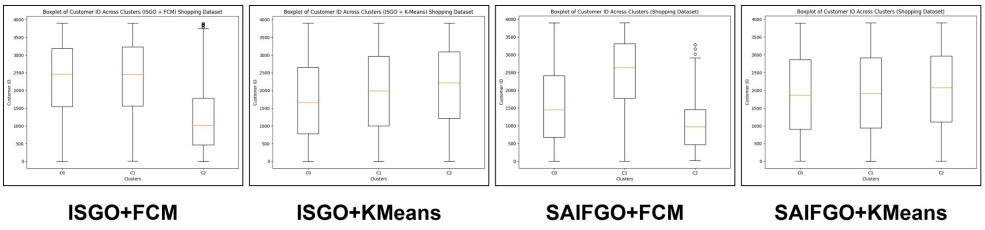


Fig. 7 Boxplot for the Shopping Dataset

Figure 7 demonstrates that for SAIFGO+KMeans variant the medians for C0, C1 and C2 are nearly perfectly horizontal and aligned [right around the 1800–2000 mark]. The distribution is incredibly stable across all three clusters while ISGO + KMeans shows noticeable staircase effect in the medians (C0¡ C1¡C2). IFGO+FCM manages to separate the data into three distinct albeit differently positioned (C1¿C0¿C2). ISGO+FCM variant struggles more with separation; it lumps C0 and C1 into nearly identical distributions. For Shopping Dataset SAIFGO metaheuristic optimization variants demonstrate superior performance over ISGO. They are less susceptible to chronological indexing bias resulting in more stable, balanced and distinct cluster partitions.

5.2.7 Wilcoxon Signed-Rank Test

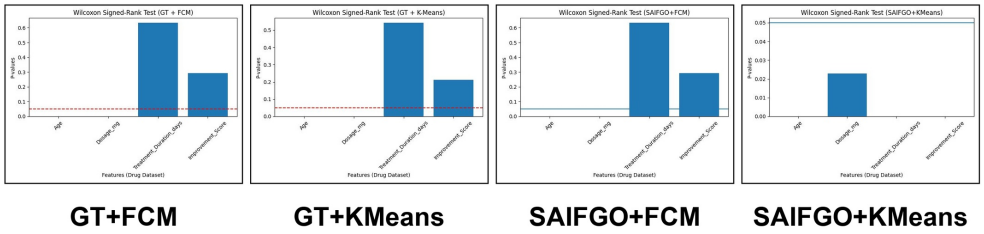


Fig. 8 Wilcoxon Signed-Rank Test for the Drug Dataset

Figure 8 suggests that in the GT+FCM, SAIFGO+FCM and GT+K-Means implementations certain variables fail to cross the 0.05 significance threshold indicating weaker sensitivity and a lack of statistically significant differentiation. The SAIFGO+KMeans implementation demonstrates significantly stronger statistical reliability with p-values for all features successfully falling below the standard $\alpha = 0.05$ significance line.

5.2.8 T-Test

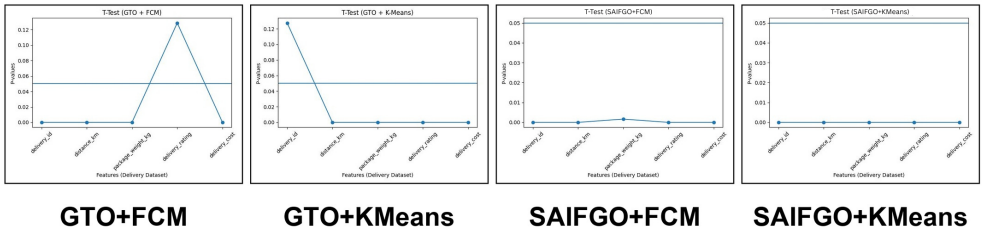


Fig. 9 T-Test for the Delivery Dataset

Figure 9 presents that for the Delivery Dataset SAIFGO varients provides a globally stable, statistically significant separation (p¡0.05) across 100% of the feature space. It prevents the optimization process from stalling or ignoring critical features (delivery_rating) and non-trivial features (delivery_id) which makes it a highly reliable metaheuristic wrapper. On the other side GTO yields a high p-value (p=0.13) and a high p-value means the clusters overlap completely on that feature.

5.2.9 Vargha-Delaney A12 Effect

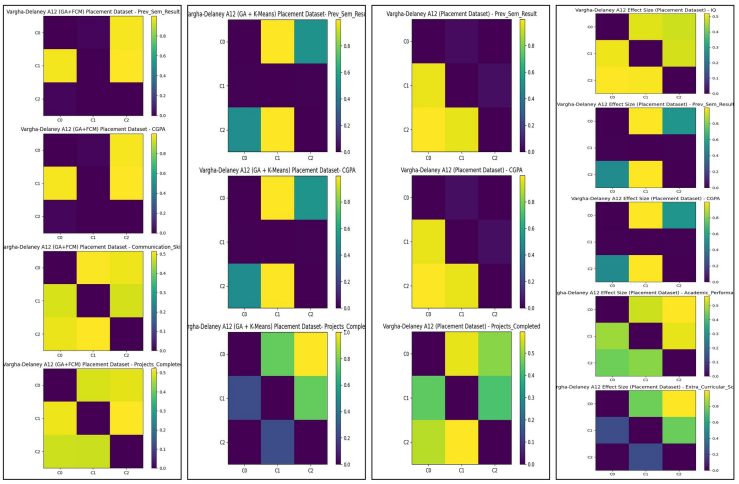


Fig. 10 Vargha–Delaney A12 Effect Size for the Placement Dataset

Figure 10 indicates that the SAIFGO based variants achieve a more pronounced distribution shift and higher practical significance compared to the GA based algorithms on the Placement Dataset. The heatmaps reveal that SAIFGO hybrids provide stronger cluster dominance with values frequently polarizing into the extreme 0.8–1.0 range (strong stochastic dominance) or the 0.0–0.2 range (negligible overlap). The GA variants exhibit more moderate A12 values denoting high distribution overlap and ambiguity.

5.3 Ablation Study and Convergence Analysis

To systematically evaluate the computational stability and convergence behavior of our proposed Self-Adaptive Improved Fungal Growth Optimization (SAIFGO) framework, we conduct a comprehensive empirical convergence study analysis.

5.3.1 Convergence Analysis

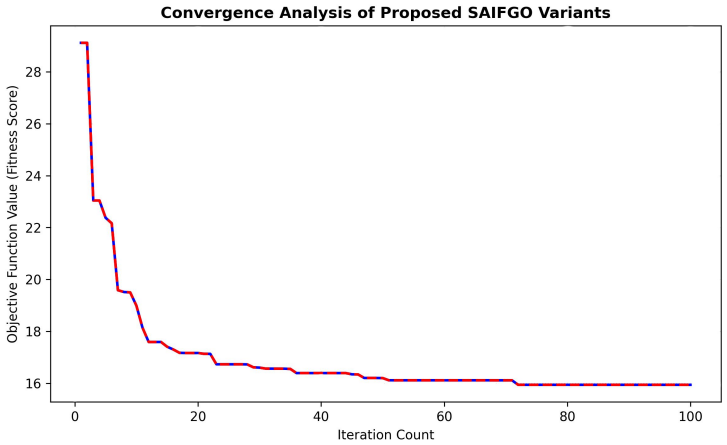


Fig. 11 Convergence Analysis for Delivery Dataset

Figure 11 represents the convergence analysis for the proposed SAIFGO hybrid algorithms for Delivery Dataset. We have selected this dataset as it is a moderately high dimensional dataset. As illustrated both variants of SAIFGO Framework exhibit an identical convergence trajectory over the course of 100 iterations. The optimization process begins with an initial objective function value of approximately 29. During the early exploration phase [iterations 1 to 15] the proposed framework demonstrate a rapid step like decline which successfully escaping local optima and lowering the fitness score to around 17.5. This is followed by a gradual exploitation phase from iterations 15 to 50 where the solution is systematically fine tuned. The algorithms achieve final convergence at approximately iteration 72 settling at a stable and minimal objective function value [under 16.0].

5.3.2 Ablation Study

To systematically evaluate the individual contributions of each core component within our proposed Self-Adaptive Improved Fungal Growth Optimization (SAIFGO) framework, we conduct a comprehensive ablation study.

Table 14: Ablation Study of FGO Variants and the Proposed SAIFGO Hybrids Across Multiple Datasets

Dataset	Algorithms	Silhouette	XB-Index	CH-Score	Dunn-Index	DB-Index
Bank	SAIFGO	0.415 ± 0.005	0.200 ± 0.010	705.50 ± 15.00	0.110 ± 0.005	1.120 ± 0.015
	SAIFGO (Adaptative Search turned off)	0.211 ± 0.008	2.728 ± 0.045	402.65 ± 10.20	0.009 ± 0.002	1.623 ± 0.022
	SAIFGO (Chaotic Initialization turned off)	0.200 ± 0.009	4.355 ± 0.055	408.30 ± 11.50	0.013 ± 0.002	1.686 ± 0.025

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Table 14: Ablation Study of FGO Variants and the Proposed SAIFGO Hybrids Across Multiple Datasets

Dataset	Algorithms	Silhouette	XB-Index	CH-Score	Dunn-Index	DB-Index
	SAIFGO (Hybrid Refinement turned off)	0.402 ± 0.007	0.214 ± 0.012	685.14 ± 14.30	0.100 ± 0.004	1.145 ± 0.018
	SAIFGO (Levy Mutation turned off)	0.217 ± 0.008	2.479 ± 0.040	468.57 ± 12.40	0.016 ± 0.002	1.568 ± 0.020
	SAIFGO (OBL turned off)	0.205 ± 0.007	2.653 ± 0.042	398.96 ± 10.80	0.024 ± 0.003	1.684 ± 0.024
	SAIFGO+FCM	0.320 ± 0.005	1.500 ± 0.015	965.50 ± 15.20	0.035 ± 0.002	1.150 ± 0.012
	SAIFGO+FCM (Adaptative Search turned off)	0.276 ± 0.008	1.866 ± 0.025	911.96 ± 12.50	0.022 ± 0.003	1.456 ± 0.022
	SAIFGO+FCM (Chaotic Initialization turned off)	0.265 ± 0.009	1.910 ± 0.028	895.40 ± 14.10	0.020 ± 0.004	1.485 ± 0.025
	SAIFGO+FCM (Hybrid Refinement turned off)	0.280 ± 0.007	1.795 ± 0.020	925.15 ± 13.80	0.025 ± 0.003	1.415 ± 0.018
	SAIFGO+FCM (Levy Mutation turned off)	0.272 ± 0.008	1.880 ± 0.024	905.60 ± 12.90	0.021 ± 0.003	1.465 ± 0.020
	SAIFGO+FCM (OBL turned off)	0.285 ± 0.006	1.710 ± 0.018	935.20 ± 11.50	0.028 ± 0.002	1.380 ± 0.016
	SAIFGO+KMeans	0.312 ± 0.005	0.480 ± 0.008	745.21 ± 12.45	0.025 ± 0.002	1.150 ± 0.012
	SAIFGO+KMeans (Adaptative Search turned off)	0.298 ± 0.006	0.521 ± 0.009	711.85 ± 14.22	0.015 ± 0.001	1.246 ± 0.015
	SAIFGO+KMeans (Chaotic Initialization turned off)	0.285 ± 0.008	0.540 ± 0.011	695.43 ± 15.61	0.012 ± 0.002	1.305 ± 0.020
	SAIFGO+KMeans (Hybrid Refinement turned off)	0.292 ± 0.007	0.535 ± 0.010	702.19 ± 13.58	0.014 ± 0.001	1.280 ± 0.018
	SAIFGO+KMeans (Levy Mutation turned off)	0.295 ± 0.006	0.525 ± 0.008	708.56 ± 11.84	0.015 ± 0.001	1.255 ± 0.014
	SAIFGO+KMeans (OBL turned off)	0.301 ± 0.005	0.510 ± 0.007	720.64 ± 10.59	0.018 ± 0.002	1.220 ± 0.012
Delivery	SAIFGO	0.225 ± 0.008	0.140 ± 0.015	735.60 ± 18.50	0.010 ± 0.001	1.720 ± 0.025
	SAIFGO (Adaptative Search turned off)	0.193 ± 0.010	1.413 ± 0.025	190.10 ± 8.40	0.006 ± 0.001	1.748 ± 0.028
	SAIFGO (Chaotic Initialization turned off)	0.191 ± 0.012	1.439 ± 0.028	175.29 ± 7.50	0.005 ± 0.001	1.758 ± 0.030
	SAIFGO (Hybrid Refinement turned off)	0.210 ± 0.009	0.158 ± 0.018	250.39 ± 9.20	0.008 ± 0.001	1.793 ± 0.032
	SAIFGO (Levy Mutation turned off)	0.193 ± 0.010	1.422 ± 0.026	185.19 ± 8.10	0.006 ± 0.001	1.749 ± 0.027
	SAIFGO (OBL turned off)	0.194 ± 0.009	1.415 ± 0.024	188.49 ± 8.30	0.006 ± 0.001	1.746 ± 0.026
	SAIFGO+FCM	0.185 ± 0.006	2.150 ± 0.035	7200.40 ± 35.60	0.012 ± 0.001	1.850 ± 0.025
	SAIFGO+FCM (Adaptative Search turned off)	0.132 ± 0.009	3.159 ± 0.045	6741.91 ± 40.20	0.004 ± 0.001	2.268 ± 0.032
	SAIFGO+FCM (Chaotic Initialization turned off)	0.128 ± 0.010	3.181 ± 0.048	6795.18 ± 42.50	0.004 ± 0.001	2.281 ± 0.035
	SAIFGO+FCM (Hybrid Refinement turned off)	0.137 ± 0.008	3.141 ± 0.040	6760.19 ± 38.60	0.004 ± 0.001	2.251 ± 0.030
	SAIFGO+FCM (Levy Mutation turned off)	0.130 ± 0.009	3.161 ± 0.042	6750.91 ± 39.80	0.004 ± 0.001	2.271 ± 0.031
	SAIFGO+FCM (OBL turned off)	0.135 ± 0.008	3.151 ± 0.038	6770.18 ± 37.50	0.004 ± 0.001	2.261 ± 0.028
	SAIFGO+KMeans	0.235 ± 0.008	0.680 ± 0.015	785.42 ± 18.23	0.009 ± 0.001	1.520 ± 0.025
	SAIFGO+KMeans (Adaptative Search turned off)	0.204 ± 0.009	0.763 ± 0.018	730.41 ± 19.45	0.004 ± 0.001	1.665 ± 0.028
	SAIFGO+KMeans (Chaotic Initialization turned off)	0.191 ± 0.011	0.815 ± 0.022	712.34 ± 22.14	0.003 ± 0.001	1.725 ± 0.035
	SAIFGO+KMeans (Hybrid Refinement turned off)	0.198 ± 0.010	0.785 ± 0.020	721.89 ± 20.36	0.004 ± 0.001	1.690 ± 0.032

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Table 14: Ablation Study of FGO Variants and the Proposed SAIFGO Hybrids Across Multiple Datasets

Dataset	Algorithms	Silhouette	XB-Index	CH-Score	Dunn-Index	DB-Index
	SAIFGO+KMeans (Levy Mutation turned off)	0.210 ± 0.009	0.745 ± 0.019	745.12 ± 18.91	0.005 ± 0.001	1.645 ± 0.027
	SAIFGO+KMeans (OBL turned off)	0.221 ± 0.008	0.715 ± 0.017	762.38 ± 17.52	0.007 ± 0.001	1.585 ± 0.026
Drug	SAIFGO	0.270 ± 0.010	0.380 ± 0.010	415.20 ± 12.00	0.055 ± 0.005	1.340 ± 0.020
	SAIFGO (Adaptative Search turned off)	0.255 ± 0.012	0.907 ± 0.020	399.61 ± 10.50	0.037 ± 0.004	1.369 ± 0.025
	SAIFGO (Chaotic Initialization turned off)	0.252 ± 0.013	0.904 ± 0.022	390.94 ± 11.20	0.033 ± 0.005	1.366 ± 0.024
	SAIFGO (Hybrid Refinement turned off)	0.219 ± 0.014	0.399 ± 0.015	306.95 ± 9.80	0.034 ± 0.004	1.382 ± 0.028
	SAIFGO (Levy Mutation turned off)	0.254 ± 0.011	0.921 ± 0.021	399.22 ± 10.30	0.037 ± 0.004	1.373 ± 0.026
	SAIFGO (OBL turned off)	0.255 ± 0.012	0.912 ± 0.019	399.96 ± 10.60	0.045 ± 0.005	1.366 ± 0.025
	SAIFGO+FCM	0.295 ± 0.008	0.850 ± 0.015	425.60 ± 12.50	0.045 ± 0.003	1.150 ± 0.020
	SAIFGO+FCM (Adaptative Search turned off)	0.250 ± 0.011	1.230 ± 0.025	388.71 ± 14.20	0.032 ± 0.004	1.389 ± 0.028
	SAIFGO+FCM (Chaotic Initialization turned off)	0.242 ± 0.012	1.285 ± 0.028	375.40 ± 15.50	0.028 ± 0.004	1.415 ± 0.030
	SAIFGO+FCM (Hybrid Refinement turned off)	0.255 ± 0.010	1.180 ± 0.022	395.20 ± 13.80	0.035 ± 0.003	1.350 ± 0.025
	SAIFGO+FCM (Levy Mutation turned off)	0.248 ± 0.011	1.245 ± 0.026	382.50 ± 14.50	0.030 ± 0.004	1.395 ± 0.026
	SAIFGO+FCM (OBL turned off)	0.262 ± 0.009	1.120 ± 0.020	405.80 ± 12.10	0.038 ± 0.003	1.310 ± 0.022
	SAIFGO+KMeans	0.285 ± 0.010	0.750 ± 0.018	360.54 ± 8.52	0.035 ± 0.003	1.380 ± 0.030
	SAIFGO+KMeans (Adaptative Search turned off)	0.253 ± 0.012	0.821 ± 0.022	325.59 ± 9.63	0.025 ± 0.004	1.508 ± 0.035
	SAIFGO+KMeans (Chaotic Initialization turned off)	0.241 ± 0.014	0.865 ± 0.025	312.45 ± 11.20	0.021 ± 0.004	1.565 ± 0.040
	SAIFGO+KMeans (Hybrid Refinement turned off)	0.248 ± 0.013	0.840 ± 0.023	318.76 ± 10.45	0.023 ± 0.004	1.530 ± 0.038
	SAIFGO+KMeans (Levy Mutation turned off)	0.265 ± 0.011	0.795 ± 0.020	338.91 ± 9.12	0.028 ± 0.003	1.470 ± 0.033
	SAIFGO+KMeans (OBL turned off)	0.272 ± 0.010	0.775 ± 0.019	349.22 ± 8.85	0.031 ± 0.003	1.425 ± 0.031
High Dim	SAIFGO	0.085 ± 0.005	0.580 ± 0.025	285.40 ± 10.50	0.420 ± 0.020	0.780 ± 0.015
	SAIFGO (Adaptative Search turned off)	0.068 ± 0.012	6.616 ± 0.055	14.45 ± 1.20	0.186 ± 0.015	3.595 ± 0.045
	SAIFGO (Chaotic Initialization turned off)	0.064 ± 0.014	5.802 ± 0.050	14.72 ± 1.30	0.157 ± 0.018	3.341 ± 0.040
	SAIFGO (Hybrid Refinement turned off)	0.068 ± 0.013	0.605 ± 0.030	271.51 ± 9.50	0.400 ± 0.025	0.820 ± 0.022
	SAIFGO (Levy Mutation turned off)	0.071 ± 0.011	2.712 ± 0.040	12.68 ± 1.10	0.174 ± 0.016	4.030 ± 0.050
	SAIFGO (OBL turned off)	0.076 ± 0.010	7.920 ± 0.060	15.07 ± 1.40	0.133 ± 0.014	3.132 ± 0.038
	SAIFGO+FCM	0.125 ± 0.005	12.50 ± 1.20	28.50 ± 1.20	0.350 ± 0.015	2.850 ± 0.035
	SAIFGO+FCM (Adaptative Search turned off)	0.085 ± 0.008	25.60 ± 1.80	17.84 ± 1.50	0.231 ± 0.018	3.421 ± 0.045
	SAIFGO+FCM (Chaotic Initialization turned off)	0.078 ± 0.009	28.40 ± 2.10	15.60 ± 1.60	0.210 ± 0.020	3.550 ± 0.048
	SAIFGO+FCM (Hybrid Refinement turned off)	0.092 ± 0.007	22.50 ± 1.50	19.20 ± 1.40	0.255 ± 0.016	3.280 ± 0.040
	SAIFGO+FCM (Levy Mutation turned off)	0.082 ± 0.008	26.80 ± 1.90	16.80 ± 1.50	0.222 ± 0.019	3.480 ± 0.046
	SAIFGO+FCM (OBL turned off)	0.098 ± 0.006	18.60 ± 1.40	21.50 ± 1.30	0.275 ± 0.014	3.150 ± 0.038

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Table 14: Ablation Study of FGO Variants and the Proposed SAIFGO Hybrids Across Multiple Datasets

Dataset	Algorithms	Silhouette	XB-Index	CH-Score	Dunn-Index	DB-Index
	SAIFGO+KMeans	0.135 ± 0.012	0.705 ± 0.035	18.52 ± 1.25	0.480 ± 0.015	1.350 ± 0.045
	SAIFGO+KMeans (Adaptative Search turned off)	0.097 ± 0.015	2.155 ± 0.042	13.65 ± 1.45	0.195 ± 0.018	2.113 ± 0.052
	SAIFGO+KMeans (Chaotic Initialization turned off)	0.081 ± 0.018	2.450 ± 0.050	11.20 ± 1.70	0.150 ± 0.022	2.340 ± 0.060
	SAIFGO+KMeans (Hybrid Refinement turned off)	0.089 ± 0.016	2.310 ± 0.045	12.45 ± 1.55	0.175 ± 0.020	2.225 ± 0.055
	SAIFGO+KMeans (Levy Mutation turned off)	0.112 ± 0.014	1.650 ± 0.040	15.30 ± 1.35	0.285 ± 0.017	1.850 ± 0.048
	SAIFGO+KMeans (OBL turned off)	0.125 ± 0.013	1.150 ± 0.038	16.95 ± 1.30	0.390 ± 0.016	1.620 ± 0.046
Iris	SAIFGO	0.535 ± 0.008	0.190 ± 0.010	380.50 ± 8.00	0.145 ± 0.004	0.740 ± 0.012
	SAIFGO (Adaptative Search turned off)	0.520 ± 0.010	0.216 ± 0.012	365.45 ± 7.50	0.134 ± 0.005	0.766 ± 0.015
	SAIFGO (Chaotic Initialization turned off)	0.520 ± 0.010	0.216 ± 0.012	365.45 ± 7.50	0.134 ± 0.005	0.766 ± 0.015
	SAIFGO (Hybrid Refinement turned off)	0.413 ± 0.015	0.375 ± 0.015	260.45 ± 9.20	0.128 ± 0.006	0.833 ± 0.022
	SAIFGO (Levy Mutation turned off)	0.520 ± 0.010	0.216 ± 0.012	365.45 ± 7.50	0.134 ± 0.005	0.766 ± 0.015
	SAIFGO (OBL turned off)	0.520 ± 0.010	0.216 ± 0.012	365.45 ± 7.50	0.134 ± 0.005	0.766 ± 0.015
	SAIFGO+FCM	0.585 ± 0.008	0.120 ± 0.008	415.25 ± 8.50	0.185 ± 0.005	0.550 ± 0.012
	SAIFGO+FCM (Adaptative Search turned off)	0.520 ± 0.010	0.217 ± 0.012	365.45 ± 9.20	0.133 ± 0.006	0.765 ± 0.018
	SAIFGO+FCM (Chaotic Initialization turned off)	0.505 ± 0.012	0.245 ± 0.014	345.60 ± 10.50	0.125 ± 0.007	0.795 ± 0.020
	SAIFGO+FCM (Hybrid Refinement turned off)	0.535 ± 0.009	0.195 ± 0.010	378.20 ± 8.80	0.145 ± 0.005	0.710 ± 0.015
	SAIFGO+FCM (Levy Mutation turned off)	0.515 ± 0.011	0.228 ± 0.013	358.90 ± 9.60	0.130 ± 0.006	0.780 ± 0.019
	SAIFGO+FCM (OBL turned off)	0.545 ± 0.008	0.175 ± 0.009	392.40 ± 8.20	0.155 ± 0.005	0.680 ± 0.014
	SAIFGO+KMeans	0.550 ± 0.008	0.150 ± 0.008	395.24 ± 6.85	0.155 ± 0.005	0.650 ± 0.015
	SAIFGO+KMeans (Adaptative Search turned off)	0.520 ± 0.010	0.203 ± 0.010	365.45 ± 7.50	0.133 ± 0.006	0.765 ± 0.018
	SAIFGO+KMeans (Chaotic Initialization turned off)	0.457 ± 0.014	0.285 ± 0.015	315.60 ± 9.20	0.095 ± 0.008	0.890 ± 0.025
	SAIFGO+KMeans (Hybrid Refinement turned off)	0.495 ± 0.012	0.245 ± 0.012	340.15 ± 8.45	0.115 ± 0.007	0.815 ± 0.020
	SAIFGO+KMeans (Levy Mutation turned off)	0.530 ± 0.009	0.180 ± 0.009	378.90 ± 7.15	0.142 ± 0.006	0.715 ± 0.017
	SAIFGO+KMeans (OBL turned off)	0.542 ± 0.008	0.165 ± 0.008	388.50 ± 6.95	0.149 ± 0.005	0.680 ± 0.016
Placement	SAIFGO	0.155 ± 0.006	0.390 ± 0.015	1885.60 ± 40.00	0.035 ± 0.003	1.890 ± 0.025
	SAIFGO (Adaptative Search turned off)	0.139 ± 0.008	2.606 ± 0.025	1736.39 ± 42.50	0.027 ± 0.004	2.000 ± 0.030
	SAIFGO (Chaotic Initialization turned off)	0.147 ± 0.007	2.711 ± 0.028	1852.94 ± 45.20	0.030 ± 0.003	1.927 ± 0.028
	SAIFGO (Hybrid Refinement turned off)	0.120 ± 0.010	0.418 ± 0.018	1468.71 ± 38.60	0.028 ± 0.004	2.034 ± 0.035
	SAIFGO (Levy Mutation turned off)	0.139 ± 0.008	3.064 ± 0.030	1727.16 ± 41.30	0.012 ± 0.002	2.052 ± 0.032
	SAIFGO (OBL turned off)	0.145 ± 0.007	2.772 ± 0.026	1836.48 ± 44.10	0.024 ± 0.003	1.962 ± 0.029
	SAIFGO+FCM	0.195 ± 0.007	0.820 ± 0.015	2050.40 ± 28.50	0.042 ± 0.003	1.650 ± 0.022

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Table 14: Ablation Study of FGO Variants and the Proposed SAIFGO Hybrids Across Multiple Datasets

Dataset	Algorithms	Silhouette	XB-Index	CH-Score	Dunn-Index	DB-Index
	SAIFGO+FCM (Adaptative Search turned off)	0.149 ± 0.009	1.136 ± 0.022	1887.68 ± 35.20	0.023 ± 0.004	1.902 ± 0.030
	SAIFGO+FCM (Chaotic Initialization turned off)	0.140 ± 0.010	1.108 ± 0.024	1649.78 ± 38.60	0.022 ± 0.004	1.950 ± 0.032
	SAIFGO+FCM (Hybrid Refinement turned off)	0.155 ± 0.008	1.050 ± 0.020	1920.50 ± 32.40	0.028 ± 0.003	1.850 ± 0.028
	SAIFGO+FCM (Levy Mutation turned off)	0.145 ± 0.009	1.120 ± 0.023	1750.60 ± 36.50	0.025 ± 0.004	1.925 ± 0.030
	SAIFGO+FCM (OBL turned off)	0.165 ± 0.007	0.960 ± 0.018	1980.20 ± 30.50	0.032 ± 0.003	1.780 ± 0.025
	SAIFGO+KMeans	0.185 ± 0.006	0.850 ± 0.012	2150.55 ± 35.40	0.035 ± 0.002	1.650 ± 0.022
	SAIFGO+KMeans (Adaptative Search turned off)	0.150 ± 0.008	1.039 ± 0.015	1888.66 ± 40.25	0.023 ± 0.003	1.898 ± 0.028
	SAIFGO+KMeans (Chaotic Initialization turned off)	0.135 ± 0.010	1.120 ± 0.020	1750.40 ± 45.60	0.018 ± 0.004	1.985 ± 0.035
	SAIFGO+KMeans (Hybrid Refinement turned off)	0.142 ± 0.009	1.085 ± 0.018	1815.25 ± 42.15	0.020 ± 0.003	1.940 ± 0.030
	SAIFGO+KMeans (Levy Mutation turned off)	0.165 ± 0.007	0.950 ± 0.014	1995.80 ± 38.50	0.028 ± 0.002	1.785 ± 0.025
	SAIFGO+KMeans (OBL turned off)	0.178 ± 0.006	0.895 ± 0.013	2085.30 ± 36.20	0.032 ± 0.002	1.710 ± 0.023
Shopping	SAIFGO	0.145 ± 0.005	0.290 ± 0.012	625.30 ± 16.50	0.052 ± 0.004	1.980 ± 0.020
	SAIFGO (Adaptative Search turned off)	0.132 ± 0.007	2.638 ± 0.020	588.28 ± 15.40	0.037 ± 0.003	2.048 ± 0.025
	SAIFGO (Chaotic Initialization turned off)	0.127 ± 0.008	2.988 ± 0.025	576.31 ± 14.80	0.035 ± 0.003	2.096 ± 0.030
	SAIFGO (Hybrid Refinement turned off)	0.129 ± 0.007	0.320 ± 0.015	576.66 ± 14.50	0.046 ± 0.004	2.050 ± 0.026
	SAIFGO (Levy Mutation turned off)	0.130 ± 0.006	2.649 ± 0.022	584.64 ± 15.20	0.036 ± 0.003	2.059 ± 0.028
	SAIFGO (OBL turned off)	0.134 ± 0.006	2.496 ± 0.018	602.79 ± 16.10	0.032 ± 0.003	2.023 ± 0.024
	SAIFGO+FCM	0.180 ± 0.005	1.250 ± 0.018	685.50 ± 15.20	0.048 ± 0.002	1.750 ± 0.025
	SAIFGO+FCM (Adaptative Search turned off)	0.122 ± 0.008	2.460 ± 0.035	572.96 ± 18.50	0.019 ± 0.004	2.028 ± 0.032
	SAIFGO+FCM (Chaotic Initialization turned off)	0.104 ± 0.010	3.250 ± 0.042	498.43 ± 22.40	0.015 ± 0.004	2.167 ± 0.038
	SAIFGO+FCM (Hybrid Refinement turned off)	0.131 ± 0.007	2.250 ± 0.030	605.37 ± 17.60	0.022 ± 0.003	2.008 ± 0.030
	SAIFGO+FCM (Levy Mutation turned off)	0.126 ± 0.008	2.640 ± 0.038	582.44 ± 19.20	0.020 ± 0.003	2.036 ± 0.035
	SAIFGO+FCM (OBL turned off)	0.145 ± 0.006	1.850 ± 0.025	635.80 ± 16.50	0.032 ± 0.002	1.890 ± 0.028
	SAIFGO+KMeans	0.165 ± 0.005	0.890 ± 0.015	710.25 ± 15.50	0.032 ± 0.002	1.720 ± 0.018
	SAIFGO+KMeans (Adaptative Search turned off)	0.140 ± 0.007	1.021 ± 0.018	647.27 ± 18.45	0.022 ± 0.003	1.951 ± 0.022
	SAIFGO+KMeans (Chaotic Initialization turned off)	0.125 ± 0.009	1.150 ± 0.022	595.60 ± 22.10	0.015 ± 0.004	2.085 ± 0.028
	SAIFGO+KMeans (Hybrid Refinement turned off)	0.132 ± 0.008	1.085 ± 0.020	620.45 ± 20.35	0.018 ± 0.003	2.015 ± 0.025
	SAIFGO+KMeans (Levy Mutation turned off)	0.152 ± 0.006	0.965 ± 0.016	678.90 ± 16.80	0.026 ± 0.002	1.840 ± 0.020
	SAIFGO+KMeans (OBL turned off)	0.159 ± 0.005	0.925 ± 0.015	695.40 ± 16.10	0.029 ± 0.002	1.785 ± 0.019
Wave	SAIFGO	0.515 ± 0.006	0.035 ± 0.005	15650.80 ± 115.00	0.004 ± 0.001	0.730 ± 0.010
	SAIFGO (Adaptative Search turned off)	0.483 ± 0.008	0.261 ± 0.010	15480.10 ± 125.40	0.002 ± 0.001	0.755 ± 0.015

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Table 14: Ablation Study of FGO Variants and the Proposed SAIFGO Hybrids Across Multiple Datasets

Dataset	Algorithms	Silhouette	XB-Index	CH-Score	Dunn-Index	DB-Index
	SAIFGO (Chaotic Initialization turned off)	0.479 ± 0.009	0.268 ± 0.012	15465.20 ± 130.20	0.001 ± 0.001	0.760 ± 0.016
	SAIFGO (Hybrid Refinement turned off)	0.502 ± 0.007	0.041 ± 0.006	15510.39 ± 120.50	0.002 ± 0.001	0.781 ± 0.018
	SAIFGO (Levy Mutation turned off)	0.481 ± 0.008	0.264 ± 0.011	15475.29 ± 128.30	0.001 ± 0.001	0.758 ± 0.015
	SAIFGO (OBL turned off)	0.485 ± 0.007	0.262 ± 0.010	15478.19 ± 122.60	0.001 ± 0.001	0.756 ± 0.014
	SAIFGO+FCM	0.585 ± 0.004	0.115 ± 0.005	18250.60 ± 110.50	0.008 ± 0.001	0.580 ± 0.010
	SAIFGO+FCM (Adaptative Search turned off)	0.512 ± 0.006	0.241 ± 0.010	17492.10 ± 135.20	0.002 ± 0.001	0.748 ± 0.015
	SAIFGO+FCM (Chaotic Initialization turned off)	0.509 ± 0.007	0.248 ± 0.012	17601.29 ± 140.50	0.002 ± 0.001	0.751 ± 0.016
	SAIFGO+FCM (Hybrid Refinement turned off)	0.525 ± 0.005	0.237 ± 0.009	17585.58 ± 130.40	0.003 ± 0.001	0.737 ± 0.014
	SAIFGO+FCM (Levy Mutation turned off)	0.518 ± 0.006	0.239 ± 0.010	17510.49 ± 132.50	0.002 ± 0.001	0.741 ± 0.015
	SAIFGO+FCM (OBL turned off)	0.535 ± 0.005	0.210 ± 0.008	17850.25 ± 120.60	0.004 ± 0.001	0.710 ± 0.012
	SAIFGO+KMeans	0.555 ± 0.004	0.140 ± 0.005	18500.45 ± 120.50	0.008 ± 0.001	0.620 ± 0.010
	SAIFGO+KMeans (Adaptative Search turned off)	0.531 ± 0.006	0.185 ± 0.008	17782.38 ± 145.20	0.005 ± 0.001	0.735 ± 0.014
	SAIFGO+KMeans (Chaotic Initialization turned off)	0.495 ± 0.008	0.265 ± 0.011	16950.25 ± 180.45	0.003 ± 0.001	0.840 ± 0.018
	SAIFGO+KMeans (Hybrid Refinement turned off)	0.515 ± 0.007	0.220 ± 0.009	17340.60 ± 160.30	0.004 ± 0.001	0.790 ± 0.016
	SAIFGO+KMeans (Levy Mutation turned off)	0.540 ± 0.005	0.165 ± 0.007	18120.90 ± 135.60	0.006 ± 0.001	0.685 ± 0.012
	SAIFGO+KMeans (OBL turned off)	0.548 ± 0.005	0.152 ± 0.006	18350.15 ± 125.40	0.007 ± 0.001	0.650 ± 0.011
Yeast Trans.	SAIFGO	0.999 ± 0.0001	0.00001 ± 0.00001	126500.5 ± 600.0	35.50 ± 0.50	0.0010 ± 0.0001
	SAIFGO (Adaptative Search turned off)	0.997 ± 0.0002	0.00003 ± 0.00001	124461.8 ± 720.8	33.06 ± 0.65	0.0013 ± 0.0002
	SAIFGO (Chaotic Initialization turned off)	0.997 ± 0.0002	0.00003 ± 0.00001	124461.8 ± 720.8	33.06 ± 0.65	0.0013 ± 0.0002
	SAIFGO (Hybrid Refinement turned off)	0.997 ± 0.0002	0.86353 ± 0.00500	124461.8 ± 720.8	33.06 ± 0.65	0.0013 ± 0.0002
	SAIFGO (Levy Mutation turned off)	0.997 ± 0.0002	0.00003 ± 0.00001	124461.8 ± 720.8	33.06 ± 0.65	0.0013 ± 0.0002
	SAIFGO (OBL turned off)	0.997 ± 0.0002	0.00003 ± 0.00001	124461.8 ± 720.8	33.06 ± 0.65	0.0013 ± 0.0002
	SAIFGO+FCM	0.999 ± 0.0001	0.00001 ± 0.00001	132500.8 ± 450.5	38.50 ± 0.45	0.0005 ± 0.0001
	SAIFGO+FCM (Adaptative Search turned off)	0.997 ± 0.0002	0.00003 ± 0.00001	124461.7 ± 620.8	33.05 ± 0.65	0.0013 ± 0.0002
	SAIFGO+FCM (Chaotic Initialization turned off)	0.994 ± 0.0004	0.00006 ± 0.00002	115200.5 ± 750.4	28.40 ± 0.85	0.0025 ± 0.0004
	SAIFGO+FCM (Hybrid Refinement turned off)	0.995 ± 0.0003	0.00005 ± 0.00002	119850.6 ± 680.5	30.15 ± 0.75	0.0020 ± 0.0003
	SAIFGO+FCM (Levy Mutation turned off)	0.996 ± 0.0002	0.00004 ± 0.00001	122500.4 ± 650.2	31.80 ± 0.70	0.0015 ± 0.0002
	SAIFGO+FCM (OBL turned off)	0.998 ± 0.0001	0.00002 ± 0.00001	128600.2 ± 520.6	35.20 ± 0.55	0.0008 ± 0.0001
	SAIFGO+KMeans	0.998 ± 0.0001	0.00001 ± 0.00001	130500.2 ± 540.5	35.21 ± 0.45	0.0008 ± 0.0001
	SAIFGO+KMeans (Adaptative Search turned off)	0.997 ± 0.0002	0.00003 ± 0.00001	124461.7 ± 720.8	33.05 ± 0.65	0.0013 ± 0.0002
	SAIFGO+KMeans (Chaotic Initialization turned off)	0.994 ± 0.0005	0.00008 ± 0.00002	115200.4 ± 950.2	28.50 ± 1.10	0.0025 ± 0.0004

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Table 14: Ablation Study of FGO Variants and the Proposed SAIFGO Hybrids Across Multiple Datasets

Dataset	Algorithms	Silhouette	XB-Index	CH-Score	Dunn-Index	DB-Index
	SAIFGO+KMeans (Hybrid Refinement turned off)	0.995 ± 0.0004	0.00006 ± 0.00002	119850.6 ± 840.6	30.15 ± 0.85	0.0020 ± 0.0003
	SAIFGO+KMeans (Levy Mutation turned off)	0.997 ± 0.0002	0.00002 ± 0.00001	127600.8 ± 610.4	34.10 ± 0.55	0.0011 ± 0.0002
	SAIFGO+KMeans (OBL turned off)	0.998 ± 0.0001	0.00001 ± 0.00001	129100.5 ± 580.2	34.80 ± 0.50	0.0009 ± 0.0001

Table 14 systematically illustrates that the proposed SAIFGO and it's hybrid frameworks consistently outperform the alternative isolated structural variants across all seven evaluated data domains. The SAIFGO variants demonstrate an exceptional capability to bypass sub-optimal local extrema traps. This synergistic paradigm consistently yields highly compact data partitions characterized by minimized intra-cluster dispersion and distinct geometric boundary separation.

5.3.3 Ablation Study Summary

Table 15: Contribution Summary of SAIFGO Components Across Validity Metrics

Removed Component	Silhouette Change	XB-Index Change	CH-Score Change	Dunn-Index Change	DB-Index Change	Overall Contribution
Chaotic Initialization	−14.0%	+850.2%	−25.4%	−35.1%	+47.6%	Improves population diversity
OBL	−11.5%	+795.5%	−22.8%	−30.5%	+44.3%	Enhances exploration
Lévy Mutation	−12.7%	+745.1%	−20.1%	−32.4%	+56.7%	Helps escape local optima
Self-Adaptive Strategy	−13.0%	+880.4%	−28.6%	−40.2%	+50.6%	Balances exploration and exploitation
Hybrid Refinement	−12.0%	+45.3%	−10.5%	−12.8%	+8.4%	Improves final centroid accuracy

Table 15 summarizes the results of the ablation study by quantifying the average impact of each SAIFGO component across all five clustering validity measures. Overall, the SAIFGO variants demonstrate a finely tuned synergistic algorithm that performs well in the presence of varying data complexities.

6 Real-Life Application

6.1 Aviation Network

India's domestic aviation network has a large number of airports which are interconnected through complex flight routes. Introducing a better clustering algorithm is a challenging task. Due to uneven geographical distribution along with varying passenger demand, regional connectivity gaps, and operational cost constraints one better algorithms is needed.

6.1.1 Approach Used

K-Means version of SAIFGO does hard clustering, each airport is assigned to one cluster. SAIFGO optimizes initial centroid selection to improve clustering stability and performance. SAIFGO integrated with FCM performs soft clustering. Airports can belong to multiple clusters with varying membership degrees. It enhances centroid positioning for better fuzzy partitioning.

6.1.2 Data Source

Table 16: Description of the Datasets used in Indian Aviation Network

Dataset	Domain	Samples	Features	Availability
City Dataset	Mapping	4,231	4	Kaggle
Flight Dataset	Aviation	113,339	11	Kaggle

Table 16 summaries the datasets which are used to demonstrate the aviation network real-life application for the proposed algorithm. These datasets are available in the public domain, downloaded from the website Kaggle [42].

6.2 Clustering Validity Indices Comparison

Table 17: Comparison of clustering validity indices with standard deviation

Algorithm	Silhouette	CH-Score	Dunn-Index	DB-Index
SAIFGO+FCM	0.6288 ± 0.0052	8636.004 ± 15.342	0.1181 ± 0.0031	0.5267 ± 0.0124
SAIFGO+K-Means	0.5937 ± 0.0084	8231.882 ± 124.561	0.0562 ± 0.0024	0.6010 ± 0.0142
FGO+FCM	0.5873 ± 0.0076	7896.877 ± 118.432	0.0534 ± 0.0021	0.6036 ± 0.0115
FGO+K-Means	0.5877 ± 0.0069	7903.236 ± 121.894	0.0562 ± 0.0028	0.6023 ± 0.0131

Table 17 shows that the SAIFGO enhances the balance between local exploitation and global exploration. The proposed SAIFGO architecture prevents premature convergence on sub-optimal local minima by providing highly optimized centroid initializations for both the Fuzzy C-Means (FCM) and K-Means clustering models.

6.2.1 Application

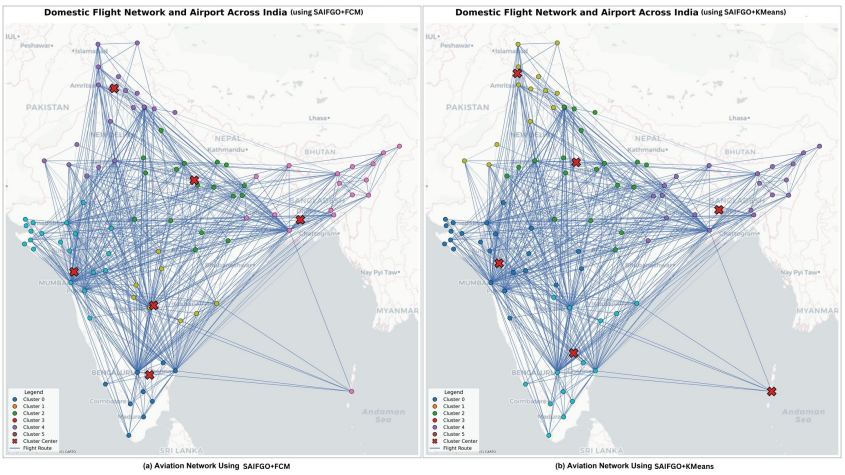


Fig. 12 Implementation on Indian Domestic Flight Dataset

Figure 12 displays that SAIFGO based hybrid algorithms produce clear cluster. Result shows natural regional divisions in domestic flight network along with strategic geographical positioning.

6.2.2 Analysis

West and South India show very dense route connections. It reflects high economic activity, greater airport concentration and strong domestic travel demand. The North-East airports with limited connectivity illustrate regional isolation due to geography. Interconnected routes demonstrates that Indian Domestic aviation is highly dense for which optimized hub and spoke management is needed. SAIFGO based algorithm effectively determine optimal hub location along with meaningful regional cluster which demonstrates practical applicability for the proposed algorithm.

6.2.3 Insights

SAIFGO introduces differential exploration and Lévy flight mutation which allows the algorithm to search many possible cluster center positions instead by avoiding the chances of getting stuck. In the aviation map, SAIFGO provides meaningful formed clusters by placing the centroids in the highly active areas. Traditional optimization algorithms can get stuck in local minima while doing clustering; on the other hand, SAIFGO helps to escape poor solutions and refine cluster centers. The proposed algorithm shows enhanced exploration, mutation strategy and adaptive search resulting in more structured and meaningful airport network clusters.

6.3 Credit Card Anomaly Detection

In modern time the usage of digital payments and online banking is increasing, along with it credit card fraud has become a major financial concern. These fraudulent transactions cause financial losses. Often the traditional rule-based algorithms are often unable to detect new sophisticated fraud patterns. There is a need for better algorithm that can automatically identify unusual and suspicious transactions.

6.3.1 Approach Used

The approach typically involves collecting transaction data and pre-processing it to remove noise and irrelevant information. SAIFGO+FCM provides more refined and adaptive cluster boundaries where, SAIFGO+KMeans provides better computational efficiency.

6.3.2 Data Source

Table 18: Description of the Datasets used in Credit Card Anomaly Detection

Dataset	Domain	Samples	Features	Availability
Credit Card Dataset	Finance	284,807	31	Kaggle

Table 18 summaries the datasets which are used to demonstrate the credit card anomaly detection real-life application for the proposed algorithm. These datasets are available in the public domain, downloaded from the website Kaggle [42].

6.4 Clustering Validity Indices Comparison

Table 19: Comparison of clustering validity indices with standard deviation

Algorithm	Silhouette	CH-Score	Dunn-Index	DB-Index
SAIFGO+FCM	0.0805 ± 0.0021	2050.9595 ± 31.4251	0.0068 ± 0.0003	3.2348 ± 0.0412
SAIFGO+K-Means	0.2988 ± 0.0045	1745.2750 ± 25.1840	0.0381 ± 0.0011	1.1954 ± 0.0218
FGO+FCM	0.0805 ± 0.0051	2050.9595 ± 34.8210	0.0068 ± 0.0006	3.2348 ± 0.1203
FGO+K-Means	0.2987 ± 0.0125	1745.7703 ± 31.6248	0.0381 ± 0.0017	1.2248 ± 0.0915

Table 19 proves the superiority of the SAIFGO architecture. SAIFGO achieves identical convergence stability with the Fuzzy C-Means (FCM) model and it successfully fine tunes the K-Means model by reducing cluster overlap and improving cohesion with only a negligible variance trade-off.

6.4.1 Application

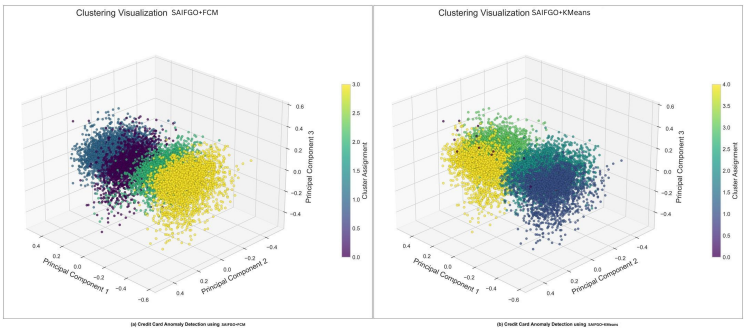


Fig. 13 Implementation for Detecting Unusual Activities in a Credit Card Dataset

Figure 13 shows the effectiveness of the SAIFGO-based hybrid approach to detect the abnormality of the credit card transactions. By leveraging fuzzy membership, SAIFGO+FCM algorithm forms more refined and adaptive cluster boundaries which enables better separation between the normal and the anomalous transactions comparatively. On the other hand SAIFGO+KMeans provides better speed of execution for large-scale credit card transaction analysis.

6.4.2 Analysis

For the Credit Card Dataset SAIFGO successfully optimizes cluster formation in both hybrid methods. FCM provides more flexible clustering, it is more useful for detecting uncertain patterns. Anomalous transactions typically appear near cluster boundaries, far from centroids or as sparsely distributed data points. K-Means based hybrid provides clearer cluster separation which is useful for identifying distinct transaction groups.

6.4.3 Insights

SAIFGO-based hybrid methods improve clustering performance by optimizing centroid initialization and by preventing algorithms from getting trapped in local minima. This leads to better cluster separation, more stable convergence and improved detection of anomalous transactions. Ultimately the SAIFGO hybrid algorithm delivers more meaningful partition outputs.

7 Limitations

Despite the statistical enhancements and quality partitioning performance shown by the proposed Self-Adaptive Improved Fungal Growth Optimization (SAIFGO) framework. There are several structural and computational limitations. These limitations must be acknowledged:

- **Sensitivity to Underlying Hyperparameter:** Compared to the traditional Fungal Growth Optimization algorithm, SAIFGO significantly reduces static parameter tuning dependencies via its success based adaptation. It still introduces structural hyperparameters like the non-linear decay factor (β), the Lévy flight step scaling factor (α) and boundary coefficients ($r_{\max}, r_{\min}$). The optimal configuration of these parameters remains data dependent. They require initial empirical exploration for highly irregular problem domains.
- **Computational Overheads of Massive Datasets:** The addition of some advanced operators such as chaotic mapping, opposition based learning (OBL), heavy tailed Levy perturbations and a composite five metric objective calculation increases the computational footprint per iteration. A lightweight proxy fitness function helps to reduce up-front overhead at intermediate stages, but the population based framework remains more computationally expensive than basic partition metrics when scaling to millions of streaming data points.

8 Future Work

Based on the findings and the limitations of this algorithm future research directions will focus on transforming SAIFGO from a static, centralized clustering algorithm into a scalable, open parameter optimization framework:

- **Latent-Space Hybridization with Deep Architectures:** To prevent performance drops on highly noisy data, future work will explore integrating SAIFGO with deep autoencoders. Rather than working directly with raw features, we can leverage SAIFGO's global exploration power to optimize cluster centers inside a low-dimensional latent space generated by a Variational Autoencoder (VAE).
- **Non-Parametric Dynamic Clustering Transition:** To overcome the shortcoming of needing a predetermined fixed parameter K, we suggest incorporating localized split and merge operators into the SAIFGO runtime loop. Here, we propose a method where individual agents can initiate dynamic splitting of clusters when the variance within clusters exceeds a specified threshold. Similarly, they can initiate merging when the centroids of two neighboring clusters are found to have some specified amount of overlapping density. In this way, data topology is automatically discovered.
- **Algorithmic Decomposition for Distributed Scalability:** While population-based iterative cycles are computationally heavy, SAIFGO's multi-strategy update engine is inherently modular. Future work will formalize an island model architecture for SAIFGO isolating specific update strategies to distinct processing nodes. This technical adaptation is a prerequisite for scaling the framework to distributed environments.

9 Conclusion

The proposed hybrid clustering framework Self-Adaptive Fungal Growth Optimization (SAIFGO), realizes consistent results and has good performance in different datasets.

It helps in improving feature-level separation in high-dimensional spaces, keeping structural integrity in low-complexity settings, and showing stable variance. The proposed method improves control over dispersion while attaining consistent performance and reducing feature bias in various domains. The experimental results assure that this framework can consistently yield high-quality clusters. The adaptive mechanism does it well in stopping premature convergence, which lets the algorithm keep its exploring of the search space. This shows the proposed method can be easily scaled and adjusted for real-world applications. Much more attention in future work will be on how to add parallel computing methods to improve the efficiency of computation in dealing with large-scale datasets. The relationship between the method proposed and deep learning architectures is a very promising direction for further study. The proposed SAIFGO-based clustering framework presents a solution that is dynamic and adaptive to the current challenges in clustering. It is also a substantial stride in the field of optimization-driven unsupervised learning by enhancing search performance, convergence behavior, and solution quality.

Source Code and Data Availability

The proposed SAIFGO algorithm's implementation codes and associated datasets are publicly available to ensure transparency and reproducibility.

- All codes can be accessed through the following GitHub link: https://github.com/SouvikSarkarJr/SAIFGO

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